

Improved Range Avoidance for Local Circuits

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Abstract

The range avoidance problem asks, given a Boolean circuit with more outputs than inputs, to find a string outside its range. Guruswami, Lyu, and Yuan gave a deterministic polynomial-time algorithm for circuits of locality $k \geq 3$ when the output length is at least $c^k n^{(k-1)/2} \log n$. We remove the logarithmic factor for every fixed locality. For each fixed $k \geq 3$, there is a constant C_k such that range avoidance is solvable in deterministic polynomial time whenever $m \geq C_k n^{(k-1)/2}$. The result allows arbitrary local predicates, arbitrary input neighborhoods, and repeated neighborhoods. In particular, locality three admits a sufficiently large constant multiple of linear stretch, and locality four admits stretch $C_4 n^{3/2}$.

Our main technical result constructs efficiently evaluable certificates for cancellation in signed XOR systems with arbitrary supports and bounded rational coefficients. At density $M \geq C_{d,\epsilon} n^{d/2}$, the certificate dominates the maximum absolute value of the signed polynomial for every signing, while its expectation is at most $\epsilon M/2$ under every sufficiently small-biased distribution on the original signs. This allows all Fourier layers of a local circuit to use one common signing.

The proof develops a constructive use of the spectral hypergraph Moore argument of Schindler and Hastings. Large trace moments expose a small window through deterministic recovery of heavy Fourier coefficients of walk matrices. For odd XOR degree, equalizing the total weight of each original pair preserves the polynomial-to-matrix identity under pair deletion. A failed local capacity condition yields a dense forest of pairs, whose edge signs are independent under uniform original signs. Removing these forests leaves a remainder with a small trace certificate. We also obtain the tradeoff $m \geq C_k n(n/\ell)^{(k-3)/2}$ in time $\text{poly}(\text{input})(en/\ell)^{O_k(\ell)}$ for $1 \leq \ell \leq n$.

The certificates give polynomial-size one-sided distinguishers that reject the entire image and accept at least half of uniform strings, excluding local demi-bits generators at the same stretch. They also give explicit query problems requiring $\Omega_t(m^{2/(t-1)})$ bits of storage with t nonadaptive bit probes; three probes require linear storage. A separate full-degree argument gives a logarithm-free $\Omega_{t,w}(m^{2/t})$ bound for adaptive probes to fixed-size words.

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1 Introduction

Let $f : \{0, 1\}^n \rightarrow \{0, 1\}^m$ be a Boolean circuit with $m > n$. The range avoidance problem, denoted by AVOID, asks for a string $y \notin \text{Range}(f)$. Such a string exists by counting, but the counting argument does not supply a deterministic efficient construction. For a fixed integer k , the problem NC_k^0 -AVOID restricts each output coordinate to depend on at most k input variables. The local functions and their input neighborhoods may differ between coordinates. We use the explicit local representation: each coordinate is specified by its neighborhood and its truth table.

Range avoidance connects explicit constructions, pseudorandomness, and circuit lower bounds. Guruswami, Lyu, and Wang [GLW22] developed these connections for low-depth circuits and gave an efficient algorithm at locality two. Gajulapalli, Golovnev, Nagargoje, and Saraogi [GGS23] established both hardness consequences and deterministic algorithms for constant locality. Their algorithm applies when $m \geq n^{k-1}/\log n$. More recently, Guruswami, Lyu, and Yuan [GLY25, Theorem 5] obtained a deterministic polynomial-time algorithm when

$$m \geq c^k n^{(k-1)/2} \log n. \quad (1)$$

Their proof reduces range avoidance to cancellation in signed XOR systems and derandomizes the analysis of semirandom XOR refutation [GKM22, HKM23].

The logarithmic factor in Eqn. (1) is particularly visible at small locality. At locality three it separates linear stretch from $n \log n$; at locality four it separates $n^{3/2}$ from $n^{3/2} \log n$. We show that this factor can be removed for every fixed locality.

1.1 Our Results

Theorem 1 (Range avoidance). *For every fixed integer $k \geq 3$, there is a constant $C_k > 0$ and a deterministic algorithm with the following guarantee. Given an explicit k -local map $f : \{0, 1\}^n \rightarrow \{0, 1\}^m$, $n \geq 1$, with*

$$m \geq C_k n^{(k-1)/2}, \quad (2)$$

the algorithm finds a string outside $\text{Range}(f)$ in time polynomial in the input length. The local functions and neighborhoods are arbitrary, and repeated neighborhoods are allowed.

The constants and the polynomial running-time exponent may depend on k . Thus Theorem 1 is a fixed-locality result. It gives $m \geq C_3 n$ at locality three and $m \geq C_4 n^{3/2}$ at locality four. It does not give polynomial time for every $m > n$. In particular, the sufficiently large constant C_3 is essential to the statement proved here.

The technical result is a signing theorem with an expectation guarantee that remains valid under a common pseudorandom distribution. For $F \subseteq [n]$ and $x \in \{\pm 1\}^n$, write $\chi_F(x) = \prod_{v \in F} x_v$.

Theorem 2 (Certificates for signed XOR systems). *Fix an integer $d \geq 2$ and a rational $\epsilon \in (0, 1)$. There is a constant $C_{d,\epsilon} > 0$ with the following property. Given $n \geq d$, labeled sets $F_1, \dots, F_M \in \binom{[n]}{d}$, and rational coefficients $a_i \in [-1, 1]$, with*

$$M \geq C_{d,\epsilon} n^{d/2}, \quad (3)$$

a deterministic algorithm constructs a nonnegative rational-valued certificate $\mathcal{C} : \{\pm 1\}^M \rightarrow \mathbb{Q}_{\geq 0}$ and a rational $\eta > 0$ such that

$$\sup_{x \in \{\pm 1\}^n} \left| \sum_{i=1}^M a_i b_i \chi_{F_i}(x) \right| \leq \mathcal{C}(b) \quad \text{for every } b, \quad (4)$$

$$\mathbb{E}_{b \sim \mathcal{D}} \mathcal{C}(b) \leq \epsilon M / 2 \quad \text{for every fully } \eta\text{-biased distribution } \mathcal{D}. \quad (5)$$

The construction, certificate evaluation, and η^{-1} are polynomial in the explicit input size, with constants depending on d, ϵ . Labels remain distinct when their supports coincide.

A fully η -biased distribution is one under which every nonempty parity of the sign coordinates has expectation of magnitude at most η . The certificate is constructed before selecting the final signing. Its decomposition may use a fixed explicit biased space, but Eqn. (5) holds for every distribution with the stated bias. This quantifier permits certificates from several systems to be summed before one common signing is selected. The statement includes zero coefficients, negative coefficients, and arbitrary repetitions.

The level of the matrix construction gives a time–stretch tradeoff. Throughout the paper, e denotes the base of the natural logarithm.

Theorem 3 (Hierarchy tradeoff). *For each fixed $k \geq 3$, there are constants $C_k, c_k > 0$ such that, for every integer $1 \leq \ell \leq n$, range avoidance for explicit k -local maps is solvable deterministically when*

$$m \geq C_k n (n/\ell)^{(k-3)/2}, \quad (6)$$

in time

$$\text{poly}(\text{input}) (en/\ell)^{c_k \ell}. \quad (7)$$

The polynomial in Eqn. (7) can be chosen with degree depending only on k . Taking ℓ constant recovers Theorem 1; increasing ℓ reduces the required output length while increasing the running time. At ℓ comparable to n , the algorithm may enumerate all input assignments. No polynomial-time claim at nearly minimal stretch follows from this endpoint.

For $k \geq 4$, taking a sufficiently large but constant level yields any prescribed positive coefficient $\rho n^{(k-1)/2}$ in polynomial time, with exponent depending on ρ ; see Corollary 27. This is a constant-factor consequence, not a reduction of the power of n . At $k = 3$, increasing the level does not change the density.

Cryptographic and data structure consequences. Our proof certifies a large set of nonoutputs, a stronger guarantee than producing one missing string. In Theorem 26 we construct a polynomial-size deterministic test D_f that rejects every image point and accepts at least half of uniformly random strings. It also accepts at least one third of any distribution with a sufficiently small polynomial full bias. The construction is uniform in the local description of f .

This immediately breaks demi-bits security against NP/poly in the stretch regime of the main theorem. For example, every five-local Goldreich generator at $m = \rho n^2$, with fixed $\rho > 0$, has such a one-sided attack. This does not exclude its smaller-stretch instantiations or dense constant-degree polynomial generators. We use the demi-bits definition and candidate discussion of [RWZ26]. Li, Ren, and Zhong [LRZ26] give a locality-preserving projection reduction from demi-bits to hard range-avoidance instances. Applying their projection mechanism to our algorithm yields polynomial-size nondeterministic certificates for a $1 - 2^{-\Omega(m)}$ fraction of candidate outputs, after a constant-factor increase in output length (Corollary 29). This amplification concerns certificate existence under uniform candidate outputs; the deterministic, small-bias guarantee above is proved separately.

The small-bias guarantee supplies an explicit universal family of candidate nonoutputs. Applying the cell-probe connection of [KPI25, GLY25], we obtain the following.

Theorem 4 (Data structure consequences). *For every fixed $t \geq 3$, there is an explicit Boolean query problem with $m^{O_t(1)}$ rows and m queries for which every exact nonadaptive t -bit-probe representation uses $\Omega_t(m^{2/(t-1)})$ bits of storage. In particular, three nonadaptive probes require $\Omega(m)$ bits. For every fixed $t \geq 2$ and word size $w \geq 1$, there is an explicit query problem requiring $\Omega_{t,w}(m^{2/t})$ cells*

for adaptive t -probe representations with w -bit words. The latter bound also allows per-row relative Hamming error $1/4$. The entries of both query problems are computable in time polynomial in $\log m$.

The nonadaptive bound removes the logarithmic loss in the corresponding GLY reduction, at the cost of requiring a sufficiently small polynomial bias whose exponent is not optimized. The adaptive word bound keeps w fixed. For one-bit words, more specialized adaptive arguments in [GLY25] achieve stronger powers than $2/t$; we do not claim to improve those powers here.

1.2 Relation to Previous Work

Our range-avoidance reduction follows Guruswami, Lyu, and Yuan [GLY25, Section 6.1]. The change lies in the certificate construction for arbitrary-support signed XOR systems. The relevant deterministic thresholds are summarized below; locality is fixed and constants may depend on it.

Result	Locality	Sufficient output length
Gajulapalli et al. [GGNS23]	$k \geq 3$	$n^{k-1}/\log n$
Guruswami–Lyu–Yuan [GLY25]	$k \geq 3$	$c^k n^{(k-1)/2} \log n$
Theorem 1	fixed $k \geq 3$	$C_k n^{(k-1)/2}$

The matrix argument builds on the Kikuchi approach to semirandom refutation and hypergraph Moore bounds [GKM22, HKM23]. Recent proofs of the sharp hypergraph Moore bound were given by Bandeira et al. [BKNN⁺26, Theorem 1.2, version 2] and Schmidhuber and Hastings; we use the spectral construction of the latter work. The second version of Bandeira et al. covers odd as well as even uniformity. Schmidhuber and Hastings [SH26b] prove a sharp hypergraph Moore bound through a normalized matrix and a lift recording the hyperedges used an odd number of times. Their odd-uniformity argument introduces balanced transitions and reciprocal-excess weights on four row-label incidences. We use these constructions, with their roles specified in the proofs below.

An even cover is a binary linear dependence among hyperedges. Its existence alone does not give the strong-cancellation certificate in [Theorem 2](#). The present argument instead uses a local capacity condition to bound the lifted operator, and a failed trace test to find a window where that condition fails. A separate companion paper of Schmidhuber and Hastings [SH26a] gives sharp Kikuchi-hierarchy guarantees for random XOR instances. Our supports are fixed adversarially; the signs are the variables over which the certificate is averaged. We prove the required arbitrary-support statement directly.

1.3 Proof Overview

We describe the main steps for a degree- d polynomial $P(b, x) = \sum_i a_i b_i \chi_{F_i}(x)$. The signs b_i always belong to original labels.

A local spectral condition. For even d , represent each clause on a slice of the Boolean cube and normalize the resulting adjacency matrix using $\Gamma = \text{diag}(\text{row degrees}) + \bar{d}I$. A lifted state consists of a row and a parity memory of original labels. Uniform sign averaging retains precisely the walks whose final memory is empty. If each small window admits fractional representatives with capacity D per input variable, an orientation of the lifted edges has incoming weight at most $(D + 1)\ell$. This bounds the normalized lifted operator by $2\sqrt{(D + 1)\ell/\bar{d}}$.

Finding a failed window. Enumerating all logarithmic-size windows would take quasipolynomial time. We instead inspect the averaged trace. If it is large, a nonnegligible mass of empty-memory closed walks must visit a bad window. Some such state has inverse-polynomial forward mass. Entries of the signed walk matrix have nonnegative Fourier coefficients: the coefficient of a memory is exactly the mass of paths reaching that memory. A deterministic prefix search recovers the heavy coefficients using a small-biased space. This finds a bad window without constructing the full parity lift. At even degree, a failed capacity matching gives a dense block supported on that window; its contribution can be certified by enumerating the window’s input assignments.

Pairs with a common total weight. For odd d , group original clauses into equal-size bundles sharing a center, and remove the center by one Cauchy–Schwarz inequality. This produces pair terms with coefficients $a_i a_j b_i b_j$. The spectral Moore construction gives port loads at most one, but arbitrary instance weights would distort the pair polynomial. A per-pair contention estimate allows all instances of a pair to be scaled to one common total weight β . The exact polynomial-to-matrix identity then survives deletion of whole pairs.

A failed capacity condition yields a forest. After pair deletion the remaining pair graph need not be a union of cliques. We use a fractional cover condition on this graph, with capacities on residual input variables. Its failure gives a dual weighting. A forest-polytope argument and threshold integration turn that weighting into a forest containing more than D times as many edges as residual variables. On a forest, the products $b_i b_j$ are independent uniform signs under uniform original signs. The forest therefore has a small expected local certificate. Only its pairs are deleted. The original bundle normalization and diagonal term are retained throughout the iteration.

One certificate and one signing. The procedure stops when the remaining degree is small or the averaged trace is small. Rational tangent-line bounds turn forest moments and trace moments into pointwise certificates. Their Fourier coefficient masses are polynomially bounded, so their expectation bounds transfer to every sufficiently small-biased distribution. We add these expectations before selecting a sign vector. This avoids any need for independence between different forests or between the Fourier layers of the circuit.

1.4 Organization of the Paper

[Section 2](#) gives the notation, pseudorandomness tools, and weighted orientation inequality. [Section 3](#) proves deterministic extraction of a bad memory state. [Section 4](#) gives the even-degree certificate. [Section 5](#) proves the local-capacity alternative and its forest certificate. [Section 6](#) completes the odd-degree proof. [Section 7](#) assembles the signing theorem and proves the range-avoidance algorithm. [Section 8](#) proves the hierarchy tradeoff and its boundary cases. [Section 9](#) proves the one-sided tests, demi-bits consequences, and data structure lower bounds. [Section 10](#) explains what short even covers provide directly and what additional construction is required.

2 Preliminaries

2.1 Local Maps, Signed Systems, and Computation

Write $[n] = \{1, \dots, n\}$ and identify a Boolean bit u with $(-1)^u \in \{\pm 1\}$. For $F \subseteq [n]$, set $\chi_F(x) = \prod_{v \in F} x_v$, with $\chi_\emptyset = 1$. Degree in this paper means the degree of the real multilinear Fourier expansion on the sign cube. It is distinct from polynomial degree over $\mathbb{F}_2$. A labeled degree- d

system is a list $F_1, \dots, F_M \in \binom{[n]}{d}$ with rational coefficients $a_i \in [-1, 1]$. The associated polynomial is

$$P(b, x) = \sum_{i=1}^M a_i b_i \chi_{F_i}(x), \quad b \in \{\pm 1\}^M, \quad x \in \{\pm 1\}^n.$$

The labels i are distinct even when the sets F_i coincide. All sign memories and pseudorandom distributions refer to these labels.

A certificate for P is an explicitly represented function $\mathcal{C} : \{\pm 1\}^M \rightarrow \mathbb{Q}_{\geq 0}$ satisfying $\sup_x |P(b, x)| \leq \mathcal{C}(b)$ for every b . The representation consists of rational matrices, local polynomials, and a specified evaluation procedure. Polynomial-time construction includes the size of this representation and the time to evaluate it. We use the usual bit model, with rational numbers encoded by signed integer numerators and positive denominators. The accuracy parameter and the locality are fixed unless a parameterized statement explicitly says otherwise. We may delete unused input variables, so polynomial dependence on n is harmless for the explicit representation.

2.2 Small Bias and Moment Certificates

For $\mu \in \mathbb{F}_2^M$, write $\chi_\mu(b) = \prod_{i:\mu_i=1} b_i$.

Definition 5. A distribution $\mathcal{D}$ on $\{\pm 1\}^M$ is *fully η -biased* if $|\mathbb{E}_{b \sim \mathcal{D}} \chi_\mu(b)| \leq \eta$ for every nonzero $\mu \in \mathbb{F}_2^M$.

Restrictions of a fully η -biased distribution to a subset of coordinates remain fully η -biased. We use the following elementary finite-field construction; small-biased spaces were developed in [NN93, AGHP92].

Lemma 6 (Explicit small-biased spaces). *For every integer $h \geq 1$ and rational $\eta \in (0, 1)$, a fully η -biased distribution on h signs can be enumerated deterministically in time polynomial in the input length and η^{-1} . Its sample space, counted with multiplicity, has size $O((h/\eta)^2)$ and all averages of rational-valued functions are rational and can be computed exactly by enumeration.*

Proof. Choose a power of two Q with $\max\{2, h/\eta\} \leq Q < 2\max\{2, h/\eta\}$, and work over $\mathbb{F}_Q$. Enumerate all $(a, z) \in \mathbb{F}_Q^2$ and output

$$b_j = (-1)^{\text{Tr}_{\mathbb{F}_Q/\mathbb{F}_2}(az^{j-1})}, \quad 1 \leq j \leq h.$$

For a nonempty set $J \subseteq [h]$, averaging its character first over a gives the indicator that $\sum_{j \in J} z^{j-1} = 0$. This is a nonzero polynomial of degree at most $h-1$, so the character expectation is at most $(h-1)/Q \leq \eta$. A representation of $\mathbb{F}_Q$ can be found by enumerating monic degree- $\log_2 Q$ binary polynomials and testing irreducibility. Even a direct enumeration and trial-factor test is polynomial in Q , which suffices here. $\square$

Lemma 7 (Bias transfer). *Let $H(b) = \sum_\mu c_\mu \chi_\mu(b)$ and set $\|\hat{H}\|_1 = \sum_\mu |c_\mu|$. For every fully η -biased distribution $\mathcal{D}$,*

$$|\mathbb{E}_{\mathcal{D}} H - \mathbb{E}_{\mathbf{U}_M} H| \leq \eta \|\hat{H}\|_1. \quad (8)$$

Proof. The constant Fourier term has the same expectation under both distributions. Bound each nonconstant character expectation by η and apply the triangle inequality. $\square$

Lemma 8 (Even moments). *Let $\xi_1, \dots, \xi_t$ be independent uniform signs and let $c_1, \dots, c_t \in \mathbb{R}$. For every integer $r \geq 1$,*

$$\mathbb{E} \left(\sum_{i=1}^t c_i \xi_i \right)^{2r} \leq (2r-1)!! \left(\sum_i c_i^2 \right)^r \leq \left(2r \sum_i c_i^2 \right)^r. \quad (9)$$

Proof. Expand the power. Only words in which each index occurs evenly survive expectation, and their coefficient products are nonnegative. Every such word admits a pairing of equal indices. Summing over all $(2r-1)!!$ pairings overcounts the surviving words and yields the first bound. The second follows from $(2r-1)!! \leq (2r)^r$. $\square$

For an integer $p \geq 1$ and $z \geq 0$, define

$$R_p(z) := \frac{p-1+z}{p} \geq z^{1/p}. \quad (10)$$

This is the tangent-line bound for the concave function $z^{1/p}$ at $z = 1$. We also use, for $e > 0$,

$$\sqrt{z} \leq \frac{e}{2} + \frac{z}{2e}. \quad (11)$$

Both right sides are rational for rational arguments.

Lemma 9 (Scalar certificates). *Fix rational $e \in (0, 1)$. Let $P(b, x) = \sum_{i=1}^M a_i b_i \chi_{F_i}(x)$ use at most u input variables, with $|a_i| \leq 1$. If $M \geq 16e^{-2}(u+1)$, put*

$$p = 2(u+1), \quad \Phi(b) = \sum_{x \in \{\pm 1\}^u} \left(\frac{P(b, x)}{eM} \right)^p, \quad C(b) = eMR_p(\Phi(b)).$$

Then $C(b) \geq \sup_x |P(b, x)|$ for every b . For every fully η -biased distribution with $\eta \leq e^{2(u+1)}/(8 \cdot 2^u)$, one has $\mathbb{E}C \leq eM$. Evaluation takes time polynomial in the input and 2^u .

Proof. Maximum-versus-sum and Eqn. (10) prove pointwise soundness. By Lemma 8,

$$\mathbb{E}_{\mathbf{U}_M} \Phi \leq 2^u \left(\frac{2(u+1)}{e^2 M} \right)^{u+1} \leq 2^u 8^{-(u+1)} \leq \frac{1}{8}.$$

Expanding the moment in the original signs gives $\|\hat{\Phi}\|_1 \leq 2^u e^{-2(u+1)}$; reducing powers modulo two can only decrease this upper bound. Lemma 7 therefore gives $\mathbb{E}\Phi \leq 1/4$, and hence $\mathbb{E}C \leq eM$. All powers and sums are rational of polynomial bit length. $\square$

2.3 Weighted Matrices and Parity Lifts

An undirected multigraph may have parallel edges with different labels. Each undirected edge contributes to both symmetric adjacency entries. Suppose its edge weights are nonnegative and its adjacency matrix is A_+ . Write N for its number of vertices, $d(S)$ for the weighted row degree, and

$$\bar{d} = N^{-1} \sum_S d(S), \quad \Gamma = \text{diag}(d(S)) + \bar{d}I.$$

When $\bar{d} > 0$, set $K_+ = \Gamma^{-1/2} A_+ \Gamma^{-1/2}$ and $P_+ = \Gamma^{-1} A_+$. The matrix P_+ is substochastic, since its row sums are $d(S)/(d(S) + \bar{d}) \leq 1$. It is similar to K_+ , so $\|K_+\| \leq 1$.

Give each edge f a memory increment $h_f \in \mathbb{F}_2^M$. Its signed weight is its unsigned weight multiplied by $\chi_{h_f}(b)$. The parity lift has states (S, μ) and replaces an edge $S \leftrightarrow S'$ by $(S, \mu) \leftrightarrow (S', \mu + h_f)$. Memories are restricted when a proof requires a finite truncated lift. Its edges inherit the normalization from the base rows, rather than using its own degrees.

Lemma 10 (Weighted orientation bound). *Let $\mathcal{L}$ be the symmetric normalized adjacency operator of any finite subgraph of such a lift. If its unnormalized edge weights admit a fractional orientation with incoming weight at most κ at every state, then*

$$\|\mathcal{L}\| \leq 2\sqrt{\kappa/\bar{d}}. \quad (12)$$

A fractional orientation splits an edge into nonnegative pieces and orients each piece independently.

Proof. This is the weighted orientation argument of Schmidhuber–Hastings [SH26b, Lemma 2.5]; we include it to specify the normalization. Taking absolute values can only increase the absolute quadratic form, so consider a nonnegative unit vector f . An oriented piece $y \rightarrow x$ of weight w contributes $2wf(x)f(y)/\sqrt{\Gamma_x\Gamma_y}$. For any $t > 0$, this is at most

$$w \left(t \frac{f(x)^2}{\Gamma_x} + t^{-1} \frac{f(y)^2}{\Gamma_y} \right).$$

The total incoming weight is at most κ and $\Gamma_x \geq \bar{d}$. The outgoing weight at y is at most its base-row degree, which is at most Γ_y . Summing gives $\langle f, \mathcal{L}f \rangle \leq t\kappa/\bar{d} + t^{-1}$. Minimization over t proves the bound. For $\kappa = 0$ there are no positive-weight edges and the conclusion is immediate. $\square$

Lemma 11 (Trace coefficient mass). *Let A_b be obtained from A_+ by multiplying each edge weight by a sign character and a fixed real number of magnitude at most one. Set $K_b = \Gamma^{-1/2}A_b\Gamma^{-1/2}$. For every integer $s \geq 1$,*

$$\left\| \widehat{\text{tr}(K_b^{2s})} \right\|_1 \leq N, \quad \text{tr}(K_b^{2s}) \geq \|K_b\|^{2s} \geq 0. \quad (13)$$

Moreover, for rational data, $\text{tr}(K_b^{2s}) = \text{tr}((\Gamma^{-1}A_b)^{2s})$ is rational and can be evaluated without computing matrix square roots.

Proof. Expand the trace into labeled closed walks and take absolute values before collecting Fourier characters. Their total is at most $\text{tr}(K_+^{2s}) \leq N\|K_+\|^{2s} \leq N$. The second assertion follows from symmetry and the even power. The rational identity follows by similarity. $\square$

If a one-label move toggles a single original sign, an empty-memory closed walk of length $2s$ has at most s active labels at any prefix. If each move toggles two labels, the corresponding bound is $2s$: at time j the memory size is at most both $2j$ and $2(2s - j)$. These bounds concern closed walks with empty final memory; arbitrary walk prefixes need not satisfy them.

3 Deterministic Extraction of a Bad Memory State

The trace arguments used later in the paper identify a collection of closed walks that pass through an obstructed state. The underlying graph has polynomially many vertices, but a state also records the parity of the original labels encountered along the walk. Enumerating all such memories would be too expensive. We show that a lower bound on the total weight of the obstructed walks allows us to recover a suitable memory deterministically. The argument uses nonnegative Fourier coefficients and applies to directed graphs, without a reversibility assumption.

3.1 Graphs with parity memory

Let G be a directed multigraph with vertex set V , where $|V| = N$. Each edge e has a nonnegative rational weight w_e and an increment $h_e \in \mathbb{F}_2^M$. Loops and parallel edges are allowed, and parallel edges retain their individual increments. The graph is given explicitly; the same statements apply when its edges can be enumerated in polynomial time. Fix positive rational numbers γ_v , $v \in V$, satisfying

$$\gamma_v \geq \sum_{e: \text{tail}(e)=v} w_e. \quad (14)$$

For $b \in \{\pm 1\}^M$ and $\mu \in \mathbb{F}_2^M$, write $\chi_\mu(b) = \prod_{i=1}^M b_i^{\mu_i}$. Define the matrices

$$A_b[v, w] = \sum_{e: v \rightarrow w} w_e \chi_{h_e}(b), \quad P_b = \Gamma^{-1} A_b, \quad \Gamma = \text{diag}(\gamma_v : v \in V). \quad (15)$$

We use the subscript $+$ for evaluation at the all-positive sign vector. Thus P_+ is entrywise nonnegative and has row sums at most one.

The *parity lift* has states $(v, \mu) \in V \times \mathbb{F}_2^M$. An edge $e : v \rightarrow w$ gives, for every μ , a transition

$$(v, \mu) \longrightarrow (w, \mu + h_e) \quad \text{of weight } w_e / \gamma_v.$$

The weight of a lifted walk is the product of its transition weights. We do not construct the parity lift explicitly.

Fix a length $T = 2s$ and a predicate $\text{Bad}(v, \mu)$ that can be evaluated in polynomial time on an explicit state. A closed empty-memory walk starts at $(u, 0)$ and returns to $(u, 0)$ after T steps. Let Δ_{bad} denote the total weight of such walks that visit a bad state at some time in $\{0, \dots, T\}$, summed over all $u \in V$. A walk that visits several bad states is counted only once in this definition. Substochasticity gives $0 \leq \Delta_{\text{bad}} \leq N$.

Theorem 12 (Deterministic memory extraction). *In the setting above, suppose that a rational number $\delta > 0$ satisfies $\Delta_{\text{bad}} \geq \delta$. There is a deterministic algorithm that finds a bad state reachable by a positive-weight lifted walk of length at most T from some empty-memory state. Its bit complexity is polynomial in the input description length, the predicate evaluation time, T , M , and $1/\delta$.*

More precisely, set

$$\tau = \frac{\delta}{2N(T+1)}. \quad (16)$$

The algorithm can enumerate a polynomial-size list containing a bad state (v, μ) for which the total weight of j -step walks from $(u, 0)$ to (v, μ) is at least τ , for some $u \in V$ and $0 \leq j \leq T$.

The final list may also contain states whose corresponding forward weight lies between $\tau/2$ and τ . Every listed state is reachable by a positive-weight walk. This distinction is useful when an application retains only states satisfying an additional restriction on their memories.

3.2 Recovering nonnegative Fourier coefficients

We first give the enumeration procedure that will be applied to entries of powers of P_b . The procedure requires exact evaluation of a function, but does not require its Fourier expansion as input.

Lemma 13 (Positive Fourier enumeration). *Let $F : \{\pm 1\}^M \rightarrow \mathbb{Q}$ have an expansion*

$$F(b) = \sum_{\mu \in \mathbb{F}_2^M} c_\mu \chi_\mu(b), \quad c_\mu \geq 0, \quad \sum_{\mu} c_\mu \leq 1. \quad (17)$$

Given an exact evaluation algorithm for F and a rational $0 < \tau \leq 1$, one can deterministically produce a list containing every μ with $c_\mu \geq \tau$, and containing only μ with $c_\mu \geq \tau/2$. The list has size at most $2/\tau$. The algorithm uses $\text{poly}(M, 1/\tau)$ evaluations of F and polynomially many additional bit operations in these parameters and the bit lengths of τ and the evaluation outputs.

Proof. For a prefix $p \in \mathbb{F}_2^h$, define its coefficient mass by

$$C(p) = \sum_{\mu: \mu_{[h]}=p} c_\mu.$$

We estimate $C(p)$ by assigning the first h signs from a small-biased space and fixing all remaining signs to $+1$. In this recovery procedure we use the separate bias parameter

$$\eta_{\text{rec}} = \tau/4. \quad (18)$$

Let $\mathcal{Q}_h$ be the explicit η_{rec} -biased distribution on h signs supplied by [Lemma 6](#). Set

$$\hat{C}(p) = \mathbb{E}_{z \sim \mathcal{Q}_h} [\chi_p(z) F(z, 1, \dots, 1)]. \quad (19)$$

This expectation is computed exactly by enumerating the sample space. Indeed, substituting Eqn. (17) into Eqn. (19) gives

$$\hat{C}(p) = \sum_{\mu} c_\mu \mathbb{E}_{z \sim \mathcal{Q}_h} \chi_{p+\mu_{[h]}}(z).$$

The summands with $\mu_{[h]} = p$ contribute exactly $C(p)$. Every other character has expectation of absolute value at most η_{rec} . Consequently,

$$\left| \hat{C}(p) - C(p) \right| \leq \eta_{\text{rec}} \sum_{\mu} c_\mu \leq \tau/4. \quad (20)$$

This is also an instance of [Lemma 7](#).

Starting with the empty prefix, extend every retained prefix by both possible bits at the next coordinate. Retain an extension p if

$$\hat{C}(p) \geq 3\tau/4. \quad (21)$$

If $c_\mu \geq \tau$, then every prefix of μ has mass at least τ , so Eqn. (20) ensures that all these prefixes are retained. Conversely, any retained prefix has mass at least $\tau/2$. Prefix masses at a fixed depth concern disjoint sets of coefficients and sum to at most one. Thus at most $2/\tau$ prefixes are retained at each depth. At depth M , a prefix specifies an individual coefficient, proving both guarantees on the output list.

There are M depths and at most $4/\tau$ oracle calls at each depth. Each oracle call enumerates a space of size $\text{poly}(M, 1/\eta_{\text{rec}}) = \text{poly}(M, 1/\tau)$. All comparison thresholds are rational, and the expectations are exact rational averages. The claimed complexity follows. $\square$

Fixing the suffix coordinates to $+1$ in Eqn. (19) is essential to this proof: it makes the oracle estimate the sum of all nonnegative coefficients extending a prefix. No individual coefficient needs to be identified before its entire index has been specified.

3.3 Forward and suffix masses

For $u, v \in V$ and $0 \leq j \leq T$, consider the efficiently evaluable function

$$F_{u,v,j}(b) = (P_b^j)[u, v].$$

Expanding matrix multiplication and reducing characters using $\chi_\mu \chi_\nu = \chi_{\mu+\nu}$ yields

$$F_{u,v,j}(b) = \sum_{\mu \in \mathbb{F}_2^M} c_{u,v,j}(\mu) \chi_\mu(b). \quad (22)$$

Here $c_{u,v,j}(\mu)$ is exactly the total weight of j -step lifted walks from $(u, 0)$ to (v, μ) . In particular, these Fourier coefficients are nonnegative. Moreover,

$$\sum_{\mu} c_{u,v,j}(\mu) = (P_+^j)[u, v], \quad \sum_{v, \mu} c_{u,v,j}(\mu) \leq 1. \quad (23)$$

Thus [Lemma 13](#) applies to every $F_{u,v,j}$.

The following argument explains why at least one recoverable coefficient describes a bad state. Translation invariance of the memory coordinate implies that the total weight of r -step walks from (v, μ) to $(u, 0)$ equals $c_{v,u,r}(\mu)$: in both cases the required memory increment is μ . This identity does not reverse the direction of any edge. It follows that the total weight of closed empty-memory T -step walks starting at $(u, 0)$ and passing through (v, μ) at time j is

$$c_{u,v,j}(\mu) c_{v,u,T-j}(\mu). \quad (24)$$

For a fixed time j , call a visit *light* relative to its starting vertex u if $c_{u,v,j}(\mu) < \tau$. Summing [Eqn. \(24\)](#) over all light visits gives

$$\begin{aligned} & \sum_{u \in V} \sum_{\substack{v \in V, \mu \in \mathbb{F}_2^M \\ c_{u,v,j}(\mu) < \tau}} c_{u,v,j}(\mu) c_{v,u,T-j}(\mu) \\ & \leq \tau \sum_{v \in V} \sum_{u \in V, \mu \in \mathbb{F}_2^M} c_{v,u,T-j}(\mu) \leq N\tau. \end{aligned} \quad (25)$$

The last inequality is the row-sum bound in [Eqn. \(23\)](#), applied with v as the starting vertex of the suffix. In particular, no estimate on a column sum or on the ratios γ_u/γ_v is needed.

Proof of [Theorem 12](#). Consider the bad closed walks for which every bad visit is light. Each such walk has at least one light bad visit. Summing [Eqn. \(25\)](#) over the $T+1$ possible visit times therefore bounds their total weight by

$$N(T+1)\tau = \delta/2.$$

Since $\Delta_{\text{bad}} \geq \delta$, there exists a bad visit (v, μ) at a time j on a closed walk starting at $(u, 0)$ such that $c_{u,v,j}(\mu) \geq \tau$.

Apply [Lemma 13](#), with the threshold in [Eqn. \(16\)](#), to each of the $N^2(T+1)$ functions $F_{u,v,j}$. Retain the indices u, j along with each recovered memory μ and endpoint v , and evaluate $\text{Bad}(v, \mu)$ on the resulting list. The bad visit just established appears in the list, so at least one test succeeds. Every listed coefficient is at least $\tau/2 > 0$ and therefore corresponds to a reachable state.

For completeness, all evaluations and comparisons admit polynomial bit complexity. Constructing P_b uses the explicitly given rational edge weights and normalizers. A common denominator for

its entries has bit length polynomial in the graph description. Powers up to T have denominators dividing the T -th power of such a denominator, and their numerators also have polynomial bit length. They can be computed by exact rational matrix multiplication. The prefix oracles use polynomially many evaluations and exact finite averages; their bit lengths remain polynomial in the stated parameters. Finally, $1/\tau = 2N(T+1)/\delta$, so the enumeration complexity in [Lemma 13](#) has the required dependence. $\square$

3.4 Truncation in the applications

The preceding theorem allows arbitrary increments. In our applications, each increment toggles either one original label or a pair of distinct original labels. The following observation permits recovery to be followed by a memory-size filter.

Suppose every increment has Hamming weight at most a , and consider a closed empty-memory walk of length $T = 2s$. If its memory after j steps is μ_j , then both its prefix and its suffix bound that memory:

$$|\text{supp}(\mu_j)| \leq \min\{aj, a(T-j)\} \leq as. \quad (26)$$

Consequently, the heavy bad visit supplied by the proof of [Theorem 12](#) survives a filter discarding all memories larger than as . The fact that other recovered states need not lie on a closed walk causes no difficulty.

For the paired construction, $a = 2$. A state consists of an ℓ -subset S of the input variables and a memory μ of original labels. If each label has a residual set R_i of size at most d , its associated window is

$$U(S, \mu) = S \cup \bigcup_{i \in \text{supp}(\mu)} R_i.$$

After the filter in [Eqn. \(26\)](#), this window has size at most $\ell + 2ds$. Hence a polynomial-time feasibility test on an explicit window can serve as the bad-state predicate, even though enumerating all windows of this size would not generally take polynomial time. When N is polynomial in the input length, $s = O(\log N)$, and δ^{-1} is polynomial, the entire extraction procedure has polynomial bit complexity.

4 Certificates for Even-Degree XOR

We first construct certificates for even-degree polynomials. The proof combines the normalized parity lift of the spectral Moore argument with a decomposition into small dense windows. The orientation argument is adapted from Schmidhuber and Hastings [[SH26b](#), Lemmas 2.5–2.7]; the decomposition removes its local sparsity hypothesis. All random characters below are characters of the original clause labels, including when several labels have identical scopes.

Theorem 14. *Fix an even integer $d \geq 2$ and a rational $0 < \epsilon < 1$. Let $F_1, \dots, F_M$ be labeled d -subsets of $[n]$, where $n \geq d$, and let $a_1, \dots, a_M$ be rational numbers with $|a_i| \leq 1$. Write $W = \sum_i |a_i|$. There is a constant $C_{d,\epsilon}$ such that, if $W \geq C_{d,\epsilon} n^{d/2}$, a deterministic algorithm constructs a nonnegative rational-valued function Cert satisfying*

$$\sup_{x \in \{\pm 1\}^n} \left| \sum_{i=1}^M a_i b_i \chi_{F_i}(x) \right| \leq \text{Cert}(b) \quad \text{for every } b \in \{\pm 1\}^M.$$

For an explicitly specified $\eta = n^{-O_{d,\epsilon}(1)} > 0$, it also satisfies

$$\mathbb{E}_{b \sim \mathcal{D}} \text{Cert}(b) \leq \frac{41}{128} \epsilon W \quad (27)$$

for every η -biased distribution $\mathcal{D}$ on the original M signs. Construction and evaluation have polynomial bit complexity for fixed d, ϵ .

The decomposition used to construct Cert may depend on a fixed explicit biased space. The distribution $\mathcal{D}$ in Eqn. (27) is arbitrary and need not be that space. This distinction will allow different Fourier layers to share one final sign vector.

4.1 Normalized matrices and local allocations

Set $\ell = d/2$, $N = \binom{n}{\ell}$, and $c = \binom{d}{\ell}$. For each label i , let A_i be the adjacency matrix on $\Omega = \binom{[n]}{\ell}$ that joins S to $S \triangle F_i$ whenever $|S \cap F_i| = \ell$. Each undirected edge is included once, so A_i has exactly c nonzero entries. Put $w_i = |a_i|$ and discard zero-weight labels from the matrices, while retaining their positions in the original sign vector.

For a nonempty set R of positive-weight labels, write

$$W_R = \sum_{i \in R} w_i, \quad A_R^{\text{abs}} = \sum_{i \in R} w_i A_i, \quad A_{R,b} = \sum_{i \in R} a_i b_i A_i.$$

Let $d_R(S)$ be the row sum of A_R^{abs} , and define

$$\bar{d}_R = \frac{c W_R}{N}, \quad \Gamma_R = \text{diag}(d_R(S)) + \bar{d}_R I, \quad K_{R,b} = \Gamma_R^{-1/2} A_{R,b} \Gamma_R^{-1/2}. \quad (28)$$

For $z(x)_S = \chi_S(x)$, direct counting gives

$$z(x)^T A_{R,b} z(x) = c \sum_{i \in R} a_i b_i \chi_{F_i}(x), \quad z(x)^T \Gamma_R z(x) = 2c W_R. \quad (29)$$

Consequently, $2W_R \|K_{R,b}\|$ is a pointwise upper bound for the absolute value of the residual polynomial.

Fix a capacity $D > 0$ and a window $V \subseteq [n]$. An allocation in V consists of nonnegative numbers $\lambda_{i,v}$, for labels $i \in R$ with $F_i \subseteq V$ and vertices $v \in F_i$, satisfying

$$\sum_{v \in F_i} \lambda_{i,v} = w_i, \quad \sum_{i: F_i \subseteq V} \lambda_{i,v} \leq D \quad (v \in V). \quad (30)$$

Feasibility is a rational maximum-flow problem: labels have demands w_i , vertices have capacities D , and a label can send flow only to its scope. The flow condition is $\sum_{i \in J} w_i \leq D |\bigcup_{i \in J} F_i|$ for every subfamily J of the labels in V . Thus a failed flow supplies a subwindow $U \subseteq V$ such that

$$\sum_{i \in R: F_i \subseteq U} w_i > D |U|. \quad (31)$$

Indeed, take U to be the union of a violating subfamily. Both the flow test and the extraction of U take polynomial time.

Lemma 15. Fix an integer $s \geq 1$. In the parity lift truncated to memories of size at most s , retain only states whose windows admit Eqn. (30). The total weight of its closed empty-memory walks of length $2s$, summed over starting rows, is at most

$$N \left(\frac{4(D+1)\ell}{\bar{d}_R} \right)^s. \quad (32)$$

The total weight before removing any states equals $\mathbb{E}_{b \text{ uniform}} \text{tr}(K_{R,b}^{2s})$.

Proof. A lifted state is (S, P) , where $S \in \Omega$ and $P \subseteq R$ has $|P| \leq s$. A move of label i changes it to $(S \triangle F_i, P \triangle \{i\})$ and has unnormalized weight w_i . Its normalized weight is $w_i / \sqrt{\Gamma_R(S, S) \Gamma_R(T, T)}$. The window of (S, P) is

$$V(S, P) = S \cup \bigcup_{i \in P} F_i.$$

Along an edge the two windows are nested: inserting i into memory replaces the window by its union with F_i .

Fix one feasible allocation for each good window. For an edge whose endpoint windows agree, split its weight into pieces $\lambda_{i,v}$ and orient each piece toward the endpoint whose row contains v . The row toggle ensures that exactly one endpoint contains v . At a fixed state (S, P) , these pieces have total incoming weight at most $D\ell$: at each $v \in S$ they are a subset of that window's allocation, and at most one incident edge has any given original label.

Orient an edge with unequal windows toward the larger window. For such an edge entering (S, P) , choose a vertex v newly present in its window. The toggled label i was inserted into memory. Since v belongs to F_i but not to the preceding row, the row toggle places it in S . Moreover, i is the unique label in P containing v , because any other such label was already present at the other endpoint. A witness v therefore determines the incoming edge. There are at most ℓ choices of witness, and each edge weighs at most one. The total incoming weight is at most $(D+1)\ell$. The outgoing weight at a state with row S is at most $d_R(S)$. By Lemma 10, the induced good-state operator has norm at most $2\sqrt{(D+1)\ell/\bar{d}_R}$. Its return mass from the N empty-memory states is bounded by Eqn. (32).

Finally, expand $\text{tr}(K_{R,b}^{2s})$ into labeled closed walks. Uniform signs kill every word in which some label occurs oddly. In each surviving word, the product of the a_i equals the product of the w_i , since all multiplicities are even. Such a word lifts uniquely to an empty-memory closed walk. At time j its memory has size at most $\min\{j, 2s-j\} \leq s$, so truncation loses none of these words. This proves the trace identity, without any assumption about even covers. $\square$

For later use, the absolute Fourier coefficient mass of $\text{tr}(K_{R,b}^{2s})$ is at most N . Indeed, the sum of absolute walk contributions is $\text{tr}((K_R^{\text{abs}})^{2s})$, where $K_R^{\text{abs}} = \Gamma_R^{-1/2} A_R^{\text{abs}} \Gamma_R^{-1/2}$. This matrix is similar to the substochastic matrix $\Gamma_R^{-1} A_R^{\text{abs}}$, hence has norm at most one. In particular, Lemma 7 gives

$$|\mathbb{E}_{b \sim \mathcal{D}} \text{tr}(K_{R,b}^{2s}) - \mathbb{E}_{b \text{ uniform}} \text{tr}(K_{R,b}^{2s})| \leq \eta N \quad (33)$$

for every η -biased $\mathcal{D}$.

4.2 A deterministic decomposition

Choose the following parameters:

$$\begin{aligned} s &= \lceil \log_2(4N) \rceil, & L &= \min\{n, \ell + ds\}, & \zeta &= \epsilon/32, \\ D &= (64/\epsilon)^2, & \beta &= \zeta^{2s}, & \eta &= \min \left\{ \frac{\beta}{8N}, (\epsilon/64)^{2L} \right\}. \end{aligned} \quad (34)$$

It is enough to impose the explicit density condition

$$W \geq \frac{64(D+1)\ell}{\epsilon\zeta^2} \frac{N}{c}. \quad (35)$$

This follows from $W \geq C_{d,\epsilon} n^{d/2}$ after choosing the constant sufficiently large. Fix one explicit η -biased space $\mathcal{Q}$ on the original M coordinates, as in [Lemma 6](#).

Start with all positive-weight labels in R and an empty list of blocks. Perform the following steps until a remainder is recorded.

1. If $W_R \leq \epsilon W/8$, record R as a small remainder and stop.
2. Otherwise form the matrices in Eqn. (28) and compute the exact rational average

$$T_{\mathcal{Q}} = \mathbb{E}_{b \sim \mathcal{Q}} \text{tr}((\Gamma_R^{-1} A_{R,b})^{2s}).$$

If $T_{\mathcal{Q}} \leq \beta$, record R as a spectral remainder and stop.

3. If $T_{\mathcal{Q}} > \beta$, find a window U satisfying Eqn. (31), with $|U| \leq L$. Record all labels of R supported on U as a block, remove them from R , and repeat.

We justify the third step. The first test has failed, and Eqn. (35) therefore implies

$$\bar{d}_R = \frac{cW_R}{N} > \frac{8(D+1)\ell}{\zeta^2}.$$

By Eqn. (33), the uniform-sign trace is greater than $7\beta/8$. On the other hand, [Lemma 15](#) bounds the mass of empty-memory closed walks staying in good states by

$$N \left(\frac{\zeta^2}{2} \right)^s \leq \frac{\beta}{4}.$$

The mass visiting a bad state is consequently greater than $5\beta/8$. Apply [Theorem 12](#), with walk length $2s$ and $\delta = \beta/2$, to the nonnegative labeled transition matrix $\Gamma_R^{-1} A_R^{\text{abs}}$. Its memory increment for label i is the i th original coordinate. Closed transition weights agree with the symmetric normalized weights by telescoping the diagonal factors. The bad predicate first rejects memories larger than s and then tests allocation feasibility in their windows. Every relevant closed walk has memory at most s , so this filter loses none of the promised bad mass. The extracted window has size at most L , and its failed flow supplies the required subwindow U .

Each iteration removes at least one positive-weight label. There are therefore at most M iterations, and the recorded blocks are disjoint in original labels. Their windows may overlap.

4.3 Pointwise certificates and their expectations

Let I_j be a recorded block with window U_j , and put $W_j = \sum_{i \in I_j} w_i$. Define its exact local certificate by

$$C_j(b) = \max_{x \in \{\pm 1\}^{U_j}} \left| \sum_{i \in I_j} a_i b_i \chi_{F_i}(x) \right|.$$

A small remainder contributes $C_R(b) = W_R$. For a spectral remainder, put

$$T_R(b) = \text{tr}((\Gamma_R^{-1} A_{R,b})^{2s}), \quad \alpha_R(b) = T_R(b)^{1/(2s)}.$$

The trace is nonnegative because the matrix is similar to a real symmetric matrix. Compute a rational $\widehat{\alpha}_R(b)$ with

$$\alpha_R(b) \leq \widehat{\alpha}_R(b) \leq \alpha_R(b) + \epsilon/256,$$

and set $C_R(b) = 2W_R\widehat{\alpha}_R(b)$. Finally, define

$$\text{Cert}(b) = \sum_j C_j(b) + C_R(b). \quad (36)$$

The local certificates are exact. For a spectral remainder, $\|K_{R,b}\| \leq \alpha_R(b)$, so Eqn. (29) proves its pointwise soundness. The triangle inequality now proves pointwise soundness of Eqn. (36) for every sign vector.

Fix an arbitrary final η -biased distribution $\mathcal{D}$. For a recorded block, set $t = |U_j| \geq 1$ and $p_b(x) = \sum_{i \in I_j} a_i b_i \chi_{F_i}(x)$. The Rademacher moment bound (Lemma 8) and $\sum_{i \in I_j} a_i^2 \leq W_j$ give

$$\mathbb{E}_{b \sim \text{uniform}} p_b(x)^{2t} \leq (2t-1)!! W_j^t \leq (2tW_j)^t.$$

As a polynomial in the original signs, $(p_b(x)/W_j)^{2t}$ has Fourier coefficient mass at most one: expanding before reducing $b_i^2 = 1$ bounds that mass by $(\sum_{i \in I_j} |a_i|/W_j)^{2t} = 1$. Thus Lemma 7 implies

$$\mathbb{E}_{b \sim \mathcal{D}} p_b(x)^{2t} \leq (2tW_j)^t + \eta W_j^{2t}.$$

Use $C_j(b)^{2t} \leq \sum_x p_b(x)^{2t}$, Jensen's inequality, and subadditivity of the $1/(2t)$ power to obtain

$$\begin{aligned} \mathbb{E}_{b \sim \mathcal{D}} C_j(b) &\leq 2\sqrt{tW_j} + \sqrt{2}\eta^{1/(2t)}W_j \\ &\leq \left(\frac{2}{\sqrt{D}} + \frac{\sqrt{2}\epsilon}{64} \right) W_j \leq \frac{\epsilon}{16} W_j. \end{aligned} \quad (37)$$

Here $W_j > Dt$ by extraction, and $t \leq L$ with Eqn. (34) gives $\eta^{1/(2t)} \leq \epsilon/64$.

A small remainder contributes at most $\epsilon W/8$. At a spectral stop, compare the final distribution to uniform and then to $\mathcal{Q}$ using Eqn. (33):

$$\mathbb{E}_{b \sim \mathcal{D}} T_R(b) \leq \beta + 2\eta N \leq \frac{5}{4}\beta.$$

Jensen's inequality gives $\mathbb{E}\alpha_R(b) \leq ((5/4)\beta)^{1/(2s)} \leq 2\zeta$, and hence

$$\mathbb{E}_{b \sim \mathcal{D}} C_R(b) \leq 4\zeta W_R + \frac{\epsilon}{128} W_R \leq \left(\frac{1}{8} + \frac{1}{128} \right) \epsilon W. \quad (38)$$

The blocks are disjoint in labels, so $\sum_j W_j \leq W$. Charging both possible remainder bounds, although only one occurs, Eqns. (37) and (38) imply

$$\mathbb{E}_{b \sim \mathcal{D}} \text{Cert}(b) \leq \left(\frac{1}{16} + \frac{1}{8} + \frac{1}{8} + \frac{1}{128} \right) \epsilon W = \frac{41}{128} \epsilon W.$$

No independence between blocks is used in summing these expectations.

4.4 Bit complexity

For fixed d, ϵ , we have $N = n^{O_d(1)}$, $s, L = O_d(\log n)$, and $\beta^{-1}, \eta^{-1} = n^{O_{d,\epsilon}(1)}$. The explicit biased space has size polynomial in M and $1/\eta$. The decomposition uses at most M iterations, and [Theorem 12](#) makes each window search polynomial in the graph input size and $1/\beta$. Every recorded local maximum can be evaluated by enumerating $2^{|U_j|} \leq 2^L$ assignments.

All matrices, capacities, and trace averages are rational. The trace is computed using $\Gamma_R^{-1} A_{R,b}$, so no matrix square roots are needed. A common denominator has polynomial bit length, and powers through $2s$ preserve a polynomial bit bound. Moreover, $T_R(b) \leq N$ by absolute domination, so $\alpha_R(b) \leq N^{1/(2s)} < 2$. Bisection on $[0, 2]$, with comparisons of rational $2s$ th powers, therefore computes the required upper approximation with polynomial bit complexity. This completes the proof of [Theorem 14](#).

5 Dense Forests from Local Capacity Violations

The odd-degree construction will delete pairs of original labels while retaining the original sign variables. A forest of such pairs is useful because its edge products are independent under uniform original signs. The first lemma obtains a dense forest from a failed fractional capacity condition. The second turns this forest into a rational certificate whose expectation is small under every sufficiently small-biased distribution on the original signs.

Lemma 16 (A forest from a capacity violation). *Let $G = (V, E)$ be a finite simple graph, let each $i \in V$ have a nonempty residual set R_i in a finite universe U , and let $D \geq 0$ be rational. Consider the system*

$$\begin{aligned} \lambda_{iv} &\geq 0 & (i \in V, v \in R_i), \\ t_i &= \sum_{v \in R_i} \lambda_{iv} & (i \in V), \\ t_i + t_j &\geq 1 & (ij \in E), \\ \sum_{i: v \in R_i} \lambda_{iv} &\leq D & (v \in U). \end{aligned} \tag{39}$$

If this system is infeasible, one can find a forest $T \subseteq E$ such that, writing $V(T)$ for its endpoints and $U_T = \bigcup_{i \in V(T)} R_i$,

$$|T| > D|U_T|. \tag{40}$$

The forest can be found deterministically in polynomial bit complexity. Moreover, adding the conditions $t_i \leq 1$ to (39) does not change its feasibility.

Proof. First suppose a feasible allocation has $t_i > 1$. Dividing all λ_{iv} at this vertex by t_i reduces its total allocation to one and decreases every capacity load. Each incident edge remains covered because its endpoint i alone now contributes one. Applying this operation at every such vertex proves the last assertion.

Farkas' lemma gives nonnegative rational edge weights y_e and vertex prices p_v , for $v \in U$, such that

$$d_i := \sum_{e \ni i} y_e \leq p_v \quad (v \in R_i), \quad \sum_{e \in E} y_e > D \sum_{v \in U} p_v. \tag{41}$$

Indeed, the coefficient inequality for the nonnegative variable λ_{iv} is precisely $d_i \leq p_v$. Since the residual sets are nonempty, any witness has $\sum_v p_v > 0$. We may therefore normalize it to $\sum_v p_v = 1$. An explicit way to find a witness is to maximize $\sum_e y_e$ subject to this normalization, nonnegativity,

and the inequalities $d_i \leq p_v$. This is a bounded rational linear program: each $d_i \leq 1$, and hence each $y_e \leq 1$. Its optimum is greater than D whenever (39) is infeasible.

For the threshold argument, replace each price by its smallest allowed value,

$$p_v = \max_{i:v \in R_i} d_i, \quad (42)$$

where an empty maximum is zero. This preserves the strict inequality in (41). If desired, rescale both y and p once more so that the price sum is one; this also preserves (42).

For $e = ij$, put $w_e = \min(d_i, d_j)$ and define

$$z_e = \begin{cases} y_e/w_e, & y_e > 0, \\ 0, & y_e = 0. \end{cases} \quad (43)$$

The denominator is positive whenever it is used. Order the vertices by increasing degree d_i , breaking ties by a fixed total order, and orient each positive- y edge towards its larger endpoint. At each vertex of positive degree, the outgoing z -weight is at most one:

$$\sum_{e \text{ directed out of } i} z_e = \frac{1}{d_i} \sum_{e \text{ directed out of } i} y_e \leq 1.$$

For every nonempty $S \subseteq V$, its largest vertex has no outgoing edge inside $G[S]$. Summing the outgoing bounds over the other vertices gives

$$\sum_{e \in E(G[S])} z_e \leq |S| - 1. \quad (44)$$

These are the graphic forest-polytope inequalities. More explicitly, for any edge set $Q \subseteq E$, apply (44) to the vertex sets of its nontrivial connected components to obtain

$$\sum_{e \in Q} z_e \leq \text{rank}(Q). \quad (45)$$

Here $\text{rank}(Q)$ denotes the number of edges in any spanning forest of (V, Q) .

We record directly the weighted consequence of these inequalities. For nonnegative edge weights w , write $E_t = \{e : w_e \geq t\}$. Kruskal's maximum-spanning-forest algorithm produces a forest T_* whose edges in E_t form a spanning forest of (V, E_t) , apart from immaterial threshold endpoints. Therefore

$$\sum_e w_e z_e = \int_0^\infty \sum_{e \in E_t} z_e dt \leq \int_0^\infty \text{rank}(E_t) dt = \sum_{e \in T_*} w_e. \quad (46)$$

This also supplies a direct justification of the needed forest-polytope property, without appealing to an integrality theorem for a linear program. Applied to the weights in (43), it gives $\sum_{e \in T_*} w_e \geq \sum_e y_e$.

For $t > 0$, let $S_t = \{i : d_i \geq t\}$ and $U_t = \bigcup_{i \in S_t} R_i$. The condition $w_{ij} \geq t$ is exactly $i, j \in S_t$, while (42) gives the second of the following identities:

$$\sum_{e \in T_*} w_e = \int_0^\infty \text{rank}(G[S_t]) dt, \quad \sum_v p_v = \int_0^\infty |U_t| dt. \quad (47)$$

Combining (41), (46), and (47), we obtain

$$\int_0^\infty (\text{rank}(G[S_t]) - D|U_t|) dt > 0.$$

Consequently some positive degree threshold satisfies $\text{rank}(G[S_t]) > D|U_t|$. Take any spanning forest of $G[S_t]$ and discard its isolated vertices. Its endpoint residual union is contained in U_t , so this forest satisfies (40).

All linear programs above have polynomially many variables and constraints and rational data of polynomial bit length. Standard rational linear programming returns an optimal solution of polynomial bit length. There are at most $|V|$ distinct positive degree thresholds, and the graph ranks and spanning forests at all of them can be computed in polynomial time. Comparisons at these rational thresholds are exact. This proves the asserted constructivity. $\square$

The preceding implication is one-way: a feasible allocation need not exclude dense forests. For example, when every residual is the same singleton, a star has a feasible allocation of capacity one by placing all allocation at its center, however many leaves the star has. We will only use infeasibility to produce a forest.

Lemma 17 (A certificate for a dense forest). *Let T be a forest whose vertices are distinct original labels in $[M]$, with residual sets R_i , and let $U_T = \bigcup_{i \in V(T)} R_i$ have size $1 \leq u \leq L$. Let $a_i \in \mathbb{Q} \cap [-1, 1]$, and put $m_T = |T|$ and*

$$Q_T(b, x) = \sum_{ij \in T} a_i a_j b_i b_j \chi_{R_i \triangle R_j}(x).$$

Suppose $\sigma \in \mathbb{Q} \cap (0, 1)$, $D \geq 32\sigma^{-2}$, and $m_T > Du$. If

$$\eta \leq \frac{\sigma^{2(L+1)}}{8 \cdot 2^L}, \quad (48)$$

then there is a nonnegative rational certificate $C_T(b)$ such that

$$\sup_{x \in \{\pm 1\}^{U_T}} |Q_T(b, x)| \leq C_T(b) \quad \text{for every } b, \quad \mathbb{E}_{b \sim \mathcal{Q}} C_T(b) \leq \sigma m_T \quad (49)$$

for every fully η -biased distribution $\mathcal{Q}$ on the original signs $b_1, \dots, b_M$. The certificate is exactly evaluable in time polynomial in the input length and 2^u .

Proof. Set $p = 2(u + 1)$ and define

$$\Phi_T(b) = \sum_{x \in \{\pm 1\}^{U_T}} \left(\frac{Q_T(b, x)}{\sigma m_T} \right)^p, \quad C_T(b) = \sigma m_T R_p(\Phi_T(b)), \quad (50)$$

where R_p is the rational envelope from (10). The exponent is even, so every summand is nonnegative. Its largest summand gives

$$\sup_x |Q_T(b, x)| \leq \sigma m_T \Phi_T(b)^{1/p} \leq \sigma m_T R_p(\Phi_T(b)) = C_T(b).$$

Under uniform independent original signs, the products $\{b_i b_j : ij \in T\}$ are mutually independent uniform signs. To see this, prescribe an arbitrary sign on every edge and choose one root sign in each connected component. Every other vertex sign is then uniquely recovered along the component's tree paths. Thus every edge-sign assignment has exactly two preimages per connected component, with all unused original signs remaining free. Repeated residual sets or repeated characters cause no change to this argument: the vertices of the forest are original labels.

For a fixed x , the independent-sign moment bound in Lemma 8, with $t = u + 1$ and coefficients of magnitude at most one, gives

$$\mathbb{E}_{\text{unif}} Q_T(b, x)^{2(u+1)} \leq (2(u + 1)m_T)^{u+1}.$$

Consequently

$$\begin{aligned}\mathbb{E}_{\text{unif}}\Phi_T &\leq 2^u \left(\frac{2(u+1)}{\sigma^2 m_T} \right)^{u+1} \\ &\leq 2^u \left(\frac{u+1}{16u} \right)^{u+1} \leq 2^u 8^{-(u+1)} \leq \frac{1}{8}.\end{aligned}\tag{51}$$

We have used $m_T > Du$, $D \geq 32\sigma^{-2}$, and $u \geq 1$.

It remains to transfer this estimate on the original sign space. For each fixed x , the sum of the absolute Fourier coefficients of $Q_T(b, x)$ as a polynomial in b is at most $\sum_{ij \in T} |a_i a_j| \leq m_T$. Multiplication of Fourier polynomials is submultiplicative for this coefficient sum; reductions using $b_i^2 = 1$ can only decrease it. Thus the Fourier coefficient sum of Φ_T is at most $2^u \sigma^{-p}$. Lemma 7 and (48) imply

$$|\mathbb{E}_{\mathcal{Q}}\Phi_T - \mathbb{E}_{\text{unif}}\Phi_T| \leq \eta 2^u \sigma^{-2(u+1)} \leq \frac{1}{8} 2^{u-L} \sigma^{2(L-u)} \leq \frac{1}{8}.\tag{52}$$

In particular, $\mathbb{E}_{\mathcal{Q}}\Phi_T \leq 1/4$. The affine formula $R_p(z) = (p-1+z)/p$ now yields

$$\mathbb{E}_{\mathcal{Q}}C_T = \sigma m_T \frac{p-1 + \mathbb{E}_{\mathcal{Q}}\Phi_T}{p} \leq \sigma m_T,$$

as required.

The construction uses only rational sums, products, and integer powers. It enumerates 2^u assignments and computes powers of degree $2(u+1)$, so the evaluation time and bit lengths are polynomial in the input length and 2^u . $\square$

Different extracted forests may share original labels. Independence across forests is neither claimed nor needed: each application of Lemma 17 bounds an expectation on the same original sign space, and linearity of expectation bounds any nonnegative weighted sum of their certificates. The final algorithm will minimize such a sum over one common biased space.

6 Certificates for Odd-Degree XOR

This section proves the signing theorem for odd degree. We retain the original clause labels throughout: two clauses with the same scope still have different sign variables. The construction first groups clauses by common centers, then represents products of residual clauses by a weighted Kikuchi matrix. A fractional local condition either bounds the appropriate memory lift or supplies a forest whose pair products can be certified separately. Equalizing the total matrix weight of each original pair makes this deletion procedure compatible with an exact polynomial identity.

Theorem 18 (Odd-degree signing certificates). *Fix an odd integer $d \geq 3$ and a rational $0 < \epsilon < 1$. There is a constant $C_{d,\epsilon}$ and a deterministic algorithm with the following property. Its input consists of M labeled sets $F_i \in \binom{[n]}{d}$ and rational coefficients $a_i \in [-1, 1]$, where*

$$M \geq C_{d,\epsilon} n^{d/2}.$$

The algorithm constructs an explicitly specified inverse-polynomial bias parameter $\eta > 0$ and an exactly evaluable nonnegative rational certificate $\mathcal{C} : \{\pm 1\}^M \rightarrow \mathbb{Q}_{\geq 0}$ such that

$$\sup_{x \in \{\pm 1\}^n} \left| \sum_{i=1}^M a_i b_i \chi_{F_i}(x) \right| \leq \mathcal{C}(b), \quad \mathbb{E}_{b \sim \mathcal{Q}} \mathcal{C}(b) \leq \frac{\epsilon M}{2}\tag{53}$$

for every b and every fully η -biased distribution $\mathcal{Q}$ on the original M signs. Construction and evaluation have polynomial bit complexity for fixed d, ϵ .

We give the proof in several steps. All constants below depend only on d, ϵ . Put

$$e = \epsilon/4, \quad B = \lceil 64e^{-2} \rceil, \quad R = \lceil 128de^{-2} \rceil, \quad \ell = 2d. \quad (54)$$

Until the final paragraph, assume $n \geq n_0 := 100d\ell$.

6.1 Bundles and their spread

For $0 \leq q \leq d-1$, define the integer threshold

$$h_q = \left\lceil BR^q \max\{1, n^{q-d/2}\} \right\rceil. \quad (55)$$

Process $q = 0, 1, \dots, d-1$ in increasing order. While a $(d-q)$ -set C is contained in at least h_q currently remaining labels, remove exactly h_q such labels as a bundle with center C . The residual of label i in this bundle is $R_i = F_i \setminus C$, of size q . The algorithm exhausts a stage before proceeding to the next one. At level q , let H_q denote the number of bundles and let $M_q = H_q h_q$. Every original label lies in at most one bundle.

Lemma 19 (Spread and the remainder). *In a level- q bundle, every nonempty residual set T of size $t \leq q$ is contained in at most $h_{q-t} - 1$ residuals. Consequently, for $q > 0$, each residual has at most qh_{q-1} overlapping partners, and*

$$\frac{qh_{q-1}}{h_q} \leq \frac{2q}{R} \leq \eta_{\text{ov}} := \frac{2d}{R} \leq \frac{e^2}{64}. \quad (56)$$

The number M_{rest} of unbundled labels satisfies

$$M_{\text{rest}} < \frac{nh_{d-1}}{d} \leq U_0 n^{d/2}, \quad U_0 := \frac{BR^{d-1} + 1}{d}. \quad (57)$$

Proof. If h_{q-t} labels in the bundle had residuals containing T , then all these labels were still present when the earlier stage $q-t$ terminated. The set $C \cup T$, of size $d - (q-t)$, would have allowed another bundle at that stage, a contradiction. Taking $t = q$ gives the bound $h_0 - 1 = B - 1$ on exact repetitions, so the argument does not require distinct scopes.

Apply the spread bound to each singleton in a fixed residual and take a union bound to count overlapping partners. Since the function $q \mapsto \max\{1, n^{q-d/2}\}$ is nondecreasing and $BR^q \geq 1$, the ceiling in (55) gives $h_{q-1}/h_q \leq 2/R$. Finally, after the last stage every input vertex belongs to fewer than h_{d-1} unbundled labels. Counting incidences gives $dM_{\text{rest}} < nh_{d-1}$, and $h_{d-1} \leq BR^{d-1}n^{d/2-1} + 1$ proves (57). $\square$

Level $q = 0$ consists of bundles of identical full scopes. Its treatment will be scalar. We now fix a positive level, suppress the subscript on $h = h_q$ and $H = H_q$, and write $M_q = Hh$.

6.2 Balanced instances and equal total pair weights

The matrix rows are indexed by $\Omega = \binom{[n]}{\ell}$, and $N = |\Omega|$. Set $a = \lfloor q/2 \rfloor$ and $c = \lceil q/2 \rceil$; these letters in this subsection are unrelated to the clause coefficients. For every unordered within-bundle pair $p = \{i, j\}$ with $R_i \cap R_j = \emptyset$, include every undirected instance

$$S \longleftrightarrow S' = S \triangle R_i \triangle R_j, \quad \{|S \cap R_i|, |S \cap R_j|\} = \{a, c\}. \quad (58)$$

Both endpoints have size ℓ . Each original pair creates exactly

$$T_q = \begin{cases} \frac{1}{2} \binom{q}{q/2}^2 \binom{n-2q}{\ell-q}, & q \text{ even}, \\ \binom{q}{\lfloor q/2 \rfloor}^2 \binom{n-2q}{\ell-q}, & q \text{ odd} \end{cases} \quad (59)$$

instances. Indeed, an even q permits one intersection pattern and an odd q permits two, after which each undirected instance has been counted twice. The instances remain labeled by their original pair even when different pairs have identical endpoints. For a nonempty positive level, the pair collection is nonempty: (56) and $1/h + \eta_{\text{ov}} < 1$ ensure that every bundle has a disjoint pair.

We use the reciprocal-excess weights of Schmidhuber–Hastings [SH26b, Section 3.2]. An instance f with endpoints S, S' and labels i, j has four ports

$$\mathcal{P}(f) = \{(S, i), (S, j), (S', i), (S', j)\}.$$

Let $c(z)$ be the number of initial instances using port z , and put

$$\delta_f = 1 + \sum_{z \in \mathcal{P}(f)} (c(z) - 1), \quad w_f = \delta_f^{-1}. \quad (60)$$

Since $\delta_f \geq c(z)$ for every port of f ,

$$\sum_{f: z \in \mathcal{P}(f)} w_f \leq 1. \quad (61)$$

Lemma 20 (Per-pair contention and equalization). *There is an explicit constant*

$$\Xi := 16BR^{d-1}(8\ell)^{d-1}$$

such that, for every original disjoint pair p ,

$$\frac{1}{T_q} \sum_{f \text{ of } p} (\delta_f - 1) \leq \Xi. \quad (62)$$

One can construct positive rational weights $\tilde{w}_f \leq w_f$ and a common rational $\beta > 0$ such that

$$\sum_{f \text{ of } p} \tilde{w}_f = \beta \quad \text{for every } p, \quad \beta \geq \frac{T_q}{1 + \Xi}. \quad (63)$$

Their port loads remain at most one. With $K_0 := 2(1 + \Xi)$, they satisfy

$$\frac{N}{\beta} \leq K_0 n^q, \quad \frac{N}{\beta h} \leq K_0 n^{d/2}. \quad (64)$$

Proof. Fix $p = \{i, j\}$. Choose one of its instances uniformly, then choose one of that instance's two endpoints uniformly and call it S . This is equivalent to choosing a balanced starting row uniformly. We first bound the expected excess at the port (S, i) . Any competing instance at this port has partner j' from the same bundle as i , with $R_{j'} \cap R_i = \emptyset$. A specified partner determines at most one instance at row S . Put $t = |R_{j'} \cap R_j|$. For a specified trace $T \subseteq R_j$ of size $t > 0$, the number of possible partners is at most h_{q-t} , by Lemma 19; for $t = 0$ it is at most h_q .

Condition on $S \cap R_i$ and $S \cap R_j$. The other $\ell - q$ row vertices form a uniformly random subset of the $n - 2q$ vertices outside $R_i \cup R_j$. If $t < a$, an eligible j' requires at least $a - t$ vertices of $R_{j'} \setminus R_j$ in this random subset. Its probability is therefore at most

$$\binom{q-t}{a-t} \left(\frac{\ell-q}{n-3q} \right)^{a-t}. \quad (65)$$

This follows by a union bound over the chosen $a - t$ vertices and the falling-factorial probability of including them. Since $n - 3q \geq n/2$, the product of (65) and the partner bound is at most

$$2BR^q 2^q (2\ell)^q (n^{-(a-t)} + n^{c-d/2}) \leq 4BR^q 2^q (2\ell)^q.$$

Here the decisive inequality is $c = \lceil q/2 \rceil \leq (d-1)/2 < d/2$. If $t \geq a$, use probability one instead: then $q - t \leq c < d/2$, so $h_{q-t} = BR^{q-t} \leq BR^q$. This includes $t = q$, and hence includes repeated residuals. Sum over at most 2^q traces and repeat the estimate after exchanging i, j . Because S was a random endpoint, twice these two expected port excesses equal the average summed excess at all four ports of an undirected instance. The resulting bound is $16BR^q (8\ell)^q \leq \Xi$. Averaging over the one or two possible intersection patterns does not change the estimate. This proves (62).

Write $W_p = \sum_{f \text{ of } p} w_f$. Cauchy–Schwarz gives

$$W_p \geq \frac{T_q^2}{\sum_{f \text{ of } p} \delta_f} \geq \frac{T_q}{1 + \Xi}.$$

Take $\beta = \min_p W_p$ and set $\tilde{w}_f = (\beta/W_p)w_f$ for each instance of p . The weights decrease and have exactly the required pair sums.

To verify the dimension bound, use

$$\frac{\binom{n-2q}{\ell-q}}{\binom{n}{\ell}} = \frac{(\ell)_q (n-\ell)_q}{(n)_{2q}} \geq \left(\frac{\ell}{4n} \right)^q.$$

The lower bound holds because $\ell \geq 2q$ and $n - \ell - q \geq n/2$. The coefficient multiplying this ratio in (59) is at least $1/2$. Thus $N/\beta \leq 2(1 + \Xi)(4n/\ell)^q \leq K_0 n^q$, since $\ell \geq 6$. Finally $h \geq \max\{1, n^{q-d/2}\}$ gives the second inequality in (64). $\square$

We freeze β and all the equalized weights. Every subsequent deletion removes an entire original pair and all of its instances. It changes no other pair weight. Thus both equalization and port capacity survive every iteration.

6.3 The comparison after pair deletion

Let E be any remaining collection of disjoint within-bundle pairs. Let $\mathcal{T}$ be any edge-disjoint family of forests whose edges, together with E , partition the initial disjoint pairs. These are forests on original clause labels; different forests may share labels. Define

$$Q_T(b, x) = \sum_{\{i,j\} \in T} a_i a_j b_i b_j \chi_{R_i \triangle R_j}(x), \quad P_q(b, x) = \sum_{\mathcal{B}} \chi_{C_{\mathcal{B}}}(x) \sum_{i \in \mathcal{B}} a_i b_i \chi_{R_i}(x).$$

Let A_b be the symmetric matrix that assigns weight $\tilde{w}_f a_i a_j b_i b_j$ to each surviving instance of $\{i, j\}$. Let A_+ use weight $\tilde{w}_f$, ignoring the coefficients as well as the signs. For $E \neq \emptyset$, put

$$d(S) = \sum_{S'} A_+[S, S'], \quad \bar{d} = \frac{2\beta|E|}{N}, \quad \Gamma = \text{diag}(d(S)) + \bar{d}I, \quad K_b = \Gamma^{-1/2} A_b \Gamma^{-1/2}. \quad (66)$$

Lemma 21 (Comparison with the original level polynomial). *With the preceding notation,*

$$\frac{\sup_x |P_q(b, x)|^2}{M_q^2} \leq \frac{1}{h} + \eta_{\text{ov}} + \frac{2}{M_q h} \sum_{T \in \mathcal{T}} \sup_x |Q_T(b, x)| + 2\|K_b\|. \quad (67)$$

The last term may instead be replaced by $2|E|/(M_q h)$, including the case $E = \emptyset$ without forming a normalized matrix.

Proof. For the row vector $z(x)_S = \chi_S(x)$, equalization gives the exact identity

$$z(x)^\top A_b z(x) = 2\beta \sum_{\{i,j\} \in E} a_i a_j b_i b_j \chi_{R_i \triangle R_j}(x). \quad (68)$$

Let O denote the initial overlapping pairs. Apply Cauchy–Schwarz once, using the original H bundles, to obtain

$$\begin{aligned} |P_q(b, x)|^2 &\leq H \sum_B \left(\sum_{i \in B} a_i b_i \chi_{R_i}(x) \right)^2 \\ &= H \sum_i a_i^2 + 2H \sum_{\{i,j\} \in O} a_i a_j b_i b_j \chi_{R_i \triangle R_j}(x) + 2H \sum_{T \in \mathcal{T}} Q_T(b, x) + \frac{H}{\beta} z(x)^\top A_b z(x). \end{aligned}$$

The first term is at most $HM_q = M_q^2/h$. By (56), $|O| \leq \eta_{\text{ov}} M_q h/2$. Moreover,

$$|z(x)^\top A_b z(x)| \leq \|K_b\| z(x)^\top \Gamma z(x) = 4\beta |E| \|K_b\|.$$

Use $|E| \leq H \binom{h}{2} \leq M_q h/2$ and divide by M_q^2 . Paying each surviving pair by one gives the alternate last term directly. In particular, the number of deleted forests does not affect the Cauchy–Schwarz factor or create additional diagonal terms. $\square$

We will stop a level if its current average degree is small. For any threshold $d_* > 0$, (64) gives

$$\bar{d} < d_* \implies \frac{2|E|}{M_q h} = \frac{N\bar{d}}{\beta M_q h} \leq K_0 d_* \frac{n^{d/2}}{M_q}. \quad (69)$$

Thus it suffices to retain only levels with a sufficiently large original count M_q ; there are at most $d - 1$ other positive levels to discard.

6.4 A fractional window condition and the memory lift

For a set $U \subseteq [n]$, take the graph on labels whose edges are the surviving pairs $\{i, j\}$ with $R_i, R_j \subseteq U$. At capacity D , call U *good* if the following linear system is feasible:

$$\lambda_{i,v} \geq 0 \quad (v \in R_i), \quad t_i := \sum_{v \in R_i} \lambda_{i,v}, \quad \sum_i \lambda_{i,v} \leq D \quad (v \in U), \quad t_i + t_j \geq 1 \quad (\{i, j\} \in E[U]). \quad (70)$$

One may impose $t_i \leq 1$ by truncating each larger vector proportionally; this preserves all pair inequalities and decreases all vertex loads. Testing goodness is a rational linear program of polynomial size.

The original-sign parity lift has states (S, μ) , where $S \in \Omega$ and $\mu \subseteq [M]$ records the active original labels. An instance of pair $\{i, j\}$ changes the state by

$$(S, \mu) \longleftrightarrow (S \triangle R_i \triangle R_j, \mu \triangle \{i, j\}).$$

Its unnormalized weight is $\tilde{w}_f$, and its normalized weight divides this by $\sqrt{\Gamma(S)\Gamma(S')}$. Associate to a state its residual window

$$W(S, \mu) = S \cup \bigcup_{i \in \mu} R_i.$$

An empty-memory closed walk of length $2s$ has at most $\min\{2j, 2(2s - j)\} \leq 2s$ active labels after j steps. All its windows therefore have size at most

$$L = \ell + 2ds. \quad (71)$$

Lemma 22 (Norm on good windows). *Restrict the normalized parity lift to states with $|\mu| \leq 2s$ and good windows. Its adjacency operator $\mathcal{L}_{\text{good}}$ satisfies*

$$\|\mathcal{L}_{\text{good}}\| \leq 2\sqrt{\frac{(D+1)\ell}{\bar{d}}}. \quad (72)$$

Proof. Fix one feasible solution to (70) for each good window. These solutions are used only to prove the bound; the algorithm does not enumerate all windows. We split and orient each lifted edge so that the total incoming unnormalized weight is at most $(D+1)\ell$.

First consider an edge whose two endpoint windows equal U , with pair $\{i, j\}$. Both residuals lie in U . Split its weight among vertices of these two residuals in fractions

$$\frac{\lambda_{i,v}}{t_i + t_j} \quad (v \in R_i), \quad \frac{\lambda_{j,v}}{t_i + t_j} \quad (v \in R_j).$$

The fractions sum to one. Since the residuals are disjoint and their union is $S \triangle S'$, each chosen vertex belongs to exactly one endpoint row. Orient the corresponding piece toward that row. At a fixed state with row S , the incoming weight witnessed by $v \in S$ is at most $\sum_i \lambda_{i,v} \leq D$: for a fixed i the relevant instances use the port (S, i) , whose total weight is at most one, and $t_i + t_j \geq 1$. Thus equal-window edges contribute at most $D\ell$.

For an unequal-window edge, choose an endpoint $x = (S, \mu)$ whose window contains a vertex absent from the other endpoint's window, breaking ties deterministically. Orient toward x and choose such a vertex v as witness. Write the other endpoint as $y = (S', \mu')$. The witness belongs to the residual of exactly one toggled label, say i , because the two residuals are disjoint. It belongs to S : otherwise the row toggle would put it in $S' \subseteq W(y)$. Label i was entered, since a removed label would put its residual in $W(y)$. Furthermore, i is the unique label in μ whose residual contains v . Any other such label would either remain in μ' or be the other toggled label, and both possibilities contradict the choice of v . Thus (x, v) determines i , and all incoming edges with witness v use the port (S, i) . They have total weight at most one. There are at most ℓ witnesses. This is the unequal-window witness argument of [SH26b, Section 3.3], applied to labeled residuals.

The lifted degree at a state with row S is at most $d(S)$. Lemma 10, with normalization $\Gamma(S) = d(S) + \bar{d}$, now gives (72). Fractionally splitting an edge is harmless for that lemma: each piece has nonnegative weight and the pieces sum to the original edge. $\square$

If U is not good, Lemma 16 constructs a forest T of surviving pairs and its residual union $U_T \subseteq U$ such that

$$|T| > D|U_T|. \quad (73)$$

All residuals are nonempty because $q > 0$. This alternative applies to the current pair graph, with no requirement that it remain a union of bundle cliques. It is the reason whole-pair deletion can be repeated.

6.5 Deterministic decomposition by a trace test

Set

$$\begin{aligned} \sigma &= \frac{e^2}{64}, & \zeta &= \frac{e^2}{128}, & D &= \lceil 32\sigma^{-2} \rceil, \\ s &= \lceil \log_2(128N) \rceil, & \beta_0 &= \zeta^{2s}, & d_* &= \frac{16(D+1)\ell}{\zeta^2}. \end{aligned} \quad (74)$$

Here β_0 is a trace threshold, distinct from the common pair mass β . Set the positive rational bias parameter to

$$\eta = \min \left\{ \frac{\beta_0}{512N}, \frac{\sigma^{2(L+1)}}{8 \cdot 2^L}, \frac{e^2}{8} \right\}, \quad L = \ell + 2ds. \quad (75)$$

For fixed d, e , its reciprocal is polynomial in n . Finally set

$$A = \left\lceil \frac{64K_0d_*}{e^2} \right\rceil \quad (76)$$

and retain a positive level only if $M_q \geq An^{d/2}$. By (69), a low-degree stop on a retained level costs at most $e^2/64$ in the squared comparison.

Fix one explicit fully η -biased distribution $\mathcal{Q}_0$ on the original M signs, using Lemma 6. On each retained level start with all its disjoint pairs and repeat:

1. If $E = \emptyset$ or $\bar{d} < d_*$, stop and use the direct pair-count bound.
2. Form the normalized matrix $K_b^{(0)}$ using instance weights $\tilde{w}_f b_i b_j$, without the $a_i a_j$ coefficients. If

$$\mathbb{E}_{b \sim \mathcal{Q}_0} \text{tr}(K_b^{(0)})^{2s} \leq \beta_0/32, \quad (77)$$

stop and retain this trace test.

3. Otherwise find a failed window, extract a forest by (73), save that forest, delete all its pairs and their instances, and repeat.

We next justify constructively the third step. The matrix $P_b = \Gamma^{-1}A_b^{(0)}$ is similar to $K_b^{(0)}$, so its traces are identical and rational. Every Fourier coefficient in an entry of P_b^j is nonnegative. Its coefficient mass is at most one, because the unsigned matrix $P_+ = \Gamma^{-1}A_+$ has row sums at most one. Likewise the nonnegative Fourier coefficient mass of $\text{tr}(K_b^{(0)})^{2s}$ is $\text{tr}(P_+^{2s}) \leq N$. Small-bias transfer therefore changes its expectation by at most ηN .

Under uniform original signs, the surviving trace terms are precisely the empty-memory closed walks of length $2s$. Failure of (77) implies that their total mass is greater than

$$\beta_0/32 - \eta N \geq 15\beta_0/512.$$

Since $\bar{d} \geq d_*$, Lemma 22 bounds the mass of those walks staying in good states by

$$N \|\mathcal{L}_{\text{good}}\|^{2s} \leq N(\zeta^2/4)^s = \beta_0 N 4^{-s} \leq \beta_0/512.$$

Here there are N empty-memory starting rows, and their length- $2s$ diagonal entries are individually bounded by the operator norm to that power. If a starting window itself is bad, walks starting there simply belong to the bad part. Thus the total mass of bad empty-memory closed walks is greater than $\delta := \beta_0/64$.

Apply Theorem 12 to the substochastic matrix P_b , with the test (70) on recovered memories. It recovers a bad state using threshold

$$\tau = \frac{\delta}{2N(2s+1)}. \quad (78)$$

Only memories with at most $2s$ labels are tested, as permitted by the empty-memory closed-walk bound. Every returned window therefore obeys $|U| \leq L$. Its inverse threshold is polynomial in n , so this step is deterministic and polynomial time. Lemma 16 then supplies the desired forest. Each iteration removes at least one original pair, and there are at most $H\binom{h}{2} \leq M_q h/2$ initial pairs. The process terminates after polynomially many iterations.

The stopping test also controls arbitrary coefficients and any final small-biased distribution. Write $K_b^{(a)}$ for the normalized matrix with the actual $a_i a_j$ coefficients. In a uniform-sign trace expansion, every surviving term contains each original label an even number of times. Its coefficient factor has the form $\prod_i a_i^{2r_i} \in [0, 1]$. Consequently

$$\mathbb{E}_{\text{unif}} \text{tr}(K_b^{(a)})^{2s} \leq \mathbb{E}_{\text{unif}} \text{tr}(K_b^{(0)})^{2s}.$$

The absolute Fourier coefficient mass of the actual trace is at most N , by domination by the unsigned path expansion. For every fully η -biased $\mathcal{Q}$, a trace stop thus implies

$$\mathbb{E}_{b \sim \mathcal{Q}} \text{tr}(K_b^{(a)})^{2s} \leq \frac{\beta_0}{32} + 2\eta N < \frac{\beta_0}{8}. \quad (79)$$

This comparison treats negative and zero coefficients without changing the pair weights, the density test, or the decomposition.

6.6 Rational certificates and the final expectation

For each extracted forest T , let $m_T = |T|$ and $u = |U_T| \leq L$. The products $b_i b_j$ on its edges are independent uniform signs under uniform original signs. Lemma 17 applies to Q_T with the parameters in (74) and (75). Explicitly, its rational potential and certificate are

$$p_T = 2(u+1), \quad \Phi_T(b) = \sum_{x \in \{\pm 1\}^{U_T}} \left(\frac{Q_T(b, x)}{\sigma m_T} \right)^{p_T}, \quad C_T(b) = \sigma m_T R_{p_T}(\Phi_T(b)), \quad (80)$$

where R_p is the rational envelope in (10). They satisfy, pointwise and for every fully η -biased $\mathcal{Q}$,

$$C_T(b) \geq \sup_x |Q_T(b, x)|, \quad \mathbb{E}_{\mathcal{Q}} C_T(b) \leq \sigma m_T. \quad (81)$$

The degree condition $m_T > Du$ is exactly (73); different forests may reuse original labels, and (81) does not assume independence between different forests.

At a trace stop define

$$\Psi_q(b) = \frac{\text{tr}(K_b^{(a)})^{2s}}{\zeta^{2s}}, \quad C_{K,q}(b) = \zeta R_{2s}(\Psi_q(b)). \quad (82)$$

The matrix is real symmetric, so its even trace is nonnegative and dominates its operator norm to the power $2s$. The envelope and (79) give

$$C_{K,q}(b) \geq \|K_b^{(a)}\|, \quad \mathbb{E}_{\mathcal{Q}} C_{K,q}(b) \leq \zeta. \quad (83)$$

The trace can be computed as $\text{tr}(\Gamma^{-1}A_b)^{2s}$; no irrational square roots are required in the algorithm.

In (67), replace each forest maximum by C_T . Use $2C_{K,q}$ at a trace stop, and $2|E|/(M_q h)$ at a low-degree stop. Call the resulting rational bound $B_q(b)$. Because the deleted forests are edge-disjoint,

$$\sum_{T \in \mathcal{T}} m_T \leq M_q h/2.$$

Together with (56), (76), and (81)–(83), this gives

$$\mathbb{E}_{\mathcal{Q}} B_q \leq \frac{e^2}{64} + \frac{e^2}{64} + \sigma + \frac{e^2}{64} = \frac{e^2}{16}. \quad (84)$$

The four terms pay respectively for the original diagonal, overlapping pairs, all deleted forests, and the final remainder. The last term is valid for both kinds of stop, since $2\zeta = e^2/64$.

Using the square-root envelope (11), define

$$C_q(b) = M_q \left(\frac{e}{2} + \frac{B_q(b)}{2e} \right). \quad (85)$$

It is pointwise at least $\sup_x |P_q(b, x)|$, and

$$\mathbb{E}_{\mathcal{Q}} C_q \leq \frac{17}{32} e M_q < e M_q. \quad (86)$$

This uses an aggregate expectation for all forests rather than a union bound over the number of extractions.

For a level-zero bundle $\mathcal{B}$, put $V_{\mathcal{B}}(b) = \sum_{i \in \mathcal{B}} a_i b_i$. Its maximum absolute contribution is exactly $|V_{\mathcal{B}}(b)|$. Use

$$\Phi_{\mathcal{B}}(b) = \left(\frac{V_{\mathcal{B}}(b)}{eB} \right)^2, \quad C_{\mathcal{B}}(b) = eB R_2(\Phi_{\mathcal{B}}(b)).$$

Its pointwise soundness follows from the envelope, and small bias gives

$$\mathbb{E}_{\mathcal{Q}} \Phi_{\mathcal{B}} \leq \frac{B + \eta B^2}{e^2 B^2} \leq \frac{1}{64} + \frac{1}{8} < \frac{1}{4}.$$

Hence $\mathbb{E}_{\mathcal{Q}} C_{\mathcal{B}} \leq eB$.

Let M_{err} be the total number of unbundled labels and labels in discarded positive levels. Equations (57) and (76) imply

$$M_{\text{err}} \leq (U_0 + (d-1)A)n^{d/2}.$$

Choose

$$C_{d,\epsilon} \geq \max \left\{ \frac{U_0 + (d-1)A}{e}, 16e^{-2}(n_0 + 1) \right\}. \quad (87)$$

Then $M_{\text{err}} \leq eM$. Sum its trivial count, all level-zero certificates, and all retained level certificates to define $\mathcal{C}$. The triangle inequality gives pointwise soundness, and the preceding expectations give

$$\mathbb{E}_{\mathcal{Q}} \mathcal{C} \leq M_{\text{err}} + e \sum_{q=0}^{d-1} M_q \leq 2eM = \epsilon M/2.$$

All terms are evaluated on the same original sign vector. In particular, the argument permits adding certificates constructed independently for different Fourier layers.

Small input lengths. If $n < n_0$, use the original polynomial $P(b, x) = \sum_i a_i b_i \chi_{F_i}(x)$ directly. Set

$$p = 2(n+1), \quad \Phi(b) = \sum_{x \in \{\pm 1\}^n} \left(\frac{P(b, x)}{eM} \right)^p, \quad \eta = \frac{e^{2(n+1)}}{8 \cdot 2^n}, \quad \mathcal{C}(b) = eMR_p(\Phi(b)).$$

The density choice (87) ensures $M \geq 16e^{-2}(n+1)$. By Lemma 8,

$$\mathbb{E}_{\text{unif}} \Phi \leq 2^n \left(\frac{2(n+1)}{e^2 M} \right)^{n+1} \leq 2^n 8^{-(n+1)} \leq \frac{1}{8}.$$

The absolute Fourier coefficient mass of Φ is at most $2^n e^{-p}$, so the displayed bias bounds the transfer error by $1/8$. Thus $\mathbb{E}_{\mathcal{Q}} \Phi \leq 1/4$ and $\mathbb{E}_{\mathcal{Q}} \mathcal{C} \leq eM < \epsilon M/2$. Pointwise soundness again follows from the even moment and its rational envelope. Enumerating 2^n inputs has constant cost depending on d, ϵ in this branch; it does not enumerate the 2^M signings.

Bit complexity. At fixed d, ϵ , there are polynomially many bundles, pairs, rows, and pair instances. All initial port loads are integers of polynomial size. The rational sums W_p and the equalized weights therefore have polynomial bit length. They are frozen during the deletion loop, so successive iterations do not recursively compound their denominators. Each trace uses a rational matrix power of order $O(\log n)$; each local certificate enumerates $2^{O(\log n)}$ assignments and uses moments of order $O(\log n)$. The explicit small-biased spaces, recovered memory lists, rational linear programs, and forest extractions all have polynomial size. Integer square-root comparisons compute the ceilings in (55) exactly. These observations establish the construction and evaluation claims in Theorem 18.

7 From Signing Certificates to Range Avoidance

We first assemble the degree- d certificate, including the conversion from coefficient mass to label count. We then apply the Fourier reduction of Guruswami, Lyu, and Yuan [GLY25, Section 6.1]. The proof selects one sign vector for all Fourier layers.

7.1 The Signing Theorem and the Elementary Degrees

Proof of Theorem 2. For odd $d \geq 3$, apply Theorem 18. Suppose d is even, and write $W = \sum_i |a_i| \leq M$. If $W \leq \epsilon M/2$, the constant certificate $\mathcal{C}(b) = W$ has both required properties; one may choose any rational $\eta \in (0, 1)$. Otherwise $W > \epsilon M/2$. Let $C_{d,\epsilon}^{\text{wt}}$ denote a sufficient density constant in Theorem 14. Choose the label-count constant so that

$$C_{d,\epsilon} \geq (2/\epsilon) C_{d,\epsilon}^{\text{wt}}.$$

The density hypothesis then gives $W > C_{d,\epsilon}^{\text{wt}} n^{d/2}$. The even-degree theorem constructs a certificate with

$$\mathbb{E} \mathcal{C}(b) \leq \frac{41}{128} \epsilon W \leq \frac{41}{128} \epsilon M < \epsilon M/2.$$

Its pointwise guarantee and bit complexity are exactly those needed. The bias parameter in either case can be decreased, if necessary, to a rational inverse-polynomial value shared with other systems. $\square$

The degrees zero and one need a different density scale from $n^{d/2}$. Their certificates are elementary.

Lemma 23 (Degree zero and degree one). *Let $n \geq 1$, let $a_i \in \mathbb{Q} \cap [-1, 1]$, and suppose either all F_i are singletons or all F_i are empty. Define*

$$\mathcal{C}_1(b) = \sum_{v=1}^n \left| \sum_{i:F_i=\{v\}} a_i b_i \right|, \quad \mathcal{C}_0(b) = \left| \sum_i a_i b_i \right|, \quad (88)$$

in the respective cases. These are exact certificates for the maximum absolute value. Under every fully η -biased distribution,

$$\mathbb{E}\mathcal{C}_1 \leq \sqrt{nM} + \sqrt{\eta}M, \quad \mathbb{E}\mathcal{C}_0 \leq \sqrt{M} + \sqrt{\eta}M. \quad (89)$$

In particular, if $M \geq 16\epsilon^{-2}n$ and $\eta \leq \epsilon^2/16$, either expectation is at most $\epsilon M/2$.

Proof. For degree one, each variable can be chosen to align with its grouped coefficient, proving exactness. The degree-zero assertion is immediate. For a group I of original labels, small bias bounds its second moment by

$$\mathbb{E} \left(\sum_{i \in I} a_i b_i \right)^2 \leq \sum_{i \in I} a_i^2 + \eta \left(\sum_{i \in I} |a_i| \right)^2.$$

Apply Cauchy–Schwarz, the inequality $\sqrt{A+B} \leq \sqrt{A} + \sqrt{B}$, and then Cauchy–Schwarz across at most n disjoint groups. Since $\sum_i a_i^2 \leq M$ and $\sum_i |a_i| \leq M$, this gives Eqn. (89). The stated parameters make each of its two summands at most $\epsilon M/4$. $\square$

7.2 Removing Parities and the Top Fourier Term

Lemma 24 (The Fourier gap). *Fix $k \geq 3$ and put $\delta = 2^{1-k}$. Let $g : \{\pm 1\}^k \rightarrow \{\pm 1\}$ be a Boolean function that is not a signed parity. There is a rational multilinear polynomial q of degree at most $k-1$ such that*

$$|g(x) - q(x)| \leq 1 - \delta \quad \text{for every } x. \quad (90)$$

Each coefficient of q has magnitude at most one, and its Fourier expansion is computable in $O_k(1)$ time.

Proof. Write $g = \sum_{A \subseteq [k]} \widehat{g}(A) \chi_A$ and put $a = \widehat{g}([k])$. The sign function $g \chi_{[k]}$ is nonconstant: otherwise g would be a signed parity. Its mean therefore lies in $[-1 + 2^{1-k}, 1 - 2^{1-k}]$. Taking $q = g - a \chi_{[k]}$ proves Eqn. (90). The remaining claims follow from $\widehat{g}(A) = 2^{-k} \sum_x g(x) \chi_A(x)$. $\square$

If a coordinate depends on fewer than k inputs and $n \geq k$, pad its neighborhood to k distinct variables and extend its truth table by ignoring the added variables. This preserves the function. It permits every coordinate to use the same set of local-position labels $[k]$, even when their neighborhoods differ.

Proof of Theorem 1. Increase C_k so that the promise implies $m > n$. If $n < k$, enumerate all 2^n inputs and their images. A missing string can be found among the first $2^n + 1$ strings of length m , in time $O_k(m)$ with a suitable dictionary. Thus assume $n \geq k$.

Identify every coordinate that is a signed parity; constant coordinates are included as parities of the empty set. Suppose there are $p > n$ such coordinates, written as $f_i(x) = s_i \chi_{T_i}(x)$, $s_i \in \{\pm 1\}$.

Their incidence vectors have a nontrivial binary dependency $\lambda \in \mathbb{F}_2^p$. Consequently every output of this submap satisfies

$$\prod_{i:\lambda_i=1} f_i(x) = \prod_{i:\lambda_i=1} s_i.$$

Choose target signs violating this equality and extend them arbitrarily to the other outputs. Binary linear algebra finds the dependency and the violating target in polynomial time.

Otherwise discard these $p \leq n$ coordinates and retain $M = m - p$ nonparity outputs. Pad and order each remaining input neighborhood as $v_{i,1}, \dots, v_{i,k}$. Apply [Lemma 24](#) to obtain

$$f_i(x) = \widehat{f}_i([k])\chi_{\{v_{i,1}, \dots, v_{i,k}\}}(x) + q_i(x), \quad |f_i(x) - q_i(x)| \leq 1 - \delta. \quad (91)$$

For every proper subset $A \subsetneq [k]$, define one layer by

$$F_{i,A} = \{v_{i,j} : j \in A\}, \quad P_A(b, x) = \sum_{i=1}^M \widehat{f}_i(A) b_i \chi_{F_{i,A}}(x).$$

There are $J = 2^k - 1$ layers and $\sum_i b_i q_i(x) = \sum_{A \subsetneq [k]} P_A(b, x)$. Zero Fourier coefficients remain as zero-weight original labels. In particular, every layer has exactly M labels on the same original sign space.

Set $\epsilon_0 = \delta/(4J)$. For $|A| \geq 2$, apply [Theorem 2](#) with parameter ϵ_0 . For $|A| \leq 1$, use [Lemma 23](#). Taking C_k sufficiently large ensures all density conditions: for $2 \leq d \leq k-1$, $n^{d/2} \leq n^{(k-1)/2}$, and $n \leq n^{(k-1)/2}$. The at most n discarded coordinates are absorbed by increasing C_k once more.

Let η_* be the minimum of the finitely many required bias parameters and $\epsilon_0^2/16$. Construct one explicit fully η_* -biased space $\mathcal{Q}_*$ on the M original output signs. Each layer supplies a pointwise certificate $\mathcal{C}_A$, and their sum satisfies

$$\mathbb{E}_{b \sim \mathcal{Q}_*} \sum_{A \subsetneq [k]} \mathcal{C}_A(b) \leq J\epsilon_0 M/2 = \delta M/8. \quad (92)$$

Enumerate this space and select a sign vector whose summed certificate is at most $\delta M/8$. In particular,

$$\sup_x \left| \sum_i b_i q_i(x) \right| < \delta M. \quad (93)$$

If an input mapped to this target on all retained coordinates, then

$$\sum_i b_i q_i(x) \geq \sum_i b_i f_i(x) - \sum_i |f_i(x) - q_i(x)| \geq M - (1 - \delta)M = \delta M,$$

contradicting Eqn. (93). Extend b arbitrarily to the discarded positions and convert back to bits. The resulting string is outside the full range.

All truth-table operations have constant cost per coordinate for fixed k . The number of layers is constant. Each certificate is constructed and evaluated in polynomial bit complexity, and η_*^{-1} is polynomial. Enumeration of $\mathcal{Q}_*$ is therefore polynomial in the explicit input length. This proves the deterministic running-time bound. $\square$

For clarity, the algorithm implemented by the proof is the following.

1. Handle bounded input size by exhaustive input evaluation. Identify signed-parity coordinates and, if there are more than n , return a target violating one binary dependency.

2. Discard the remaining signed parities, pad each neighborhood to k distinct inputs, and compute the proper Fourier layers.
3. Construct the degree-zero and degree-one certificates directly. For even higher degrees, repeatedly remove a dense window after a failed averaged trace test. For odd degrees, construct equalized pair matrices and repeatedly remove the dense forests supplied by failed window LPs. Retain every resulting local and remainder certificate.
4. Enumerate a single sufficiently small-biased space on the original output signs. Return a vector passing the summed certificate test in Eqn. (92), extended arbitrarily on discarded coordinates.

Every branch uses exact rational tests. The algorithm does not need to certify the local capacity condition on all small windows at a stopping point: the averaged trace test itself is the remainder certificate used in the proof.

8 The Hierarchy Tradeoff

We now allow the matrix level to grow with n and prove [Theorem 3](#). For $d \geq 2$ and $1 \leq \ell \leq n$, define

$$\mu_d(n, \ell) = n \left(\frac{n}{\ell} \right)^{d/2-1} = \ell \left(\frac{n}{\ell} \right)^{d/2}. \quad (94)$$

The role of the level is to increase the average matrix degree produced by each original term. The local capacity and cancellation parameters remain constants depending only on d and the desired margin.

Proposition 25. *Fix $d \geq 2$ and rational $0 < \epsilon < 1$. The certificate conclusion of [Theorem 2](#) holds under the density hypothesis $M \geq C_{d,\epsilon} \mu_d(n, \ell)$, with construction, evaluation, and inverse bias bounded by*

$$\text{poly}(\text{input}) \left(\frac{en}{\ell} \right)^{O_{d,\epsilon}(\ell)}. \quad (95)$$

The expectation guarantee continues to hold for every final distribution with the specified full bias.

8.1 The central range

Assume first that $2d \leq \ell$ and $n \geq 100d\ell$, and write $v = n/\ell$ and $N = \binom{n}{\ell}$. For $1 \leq q \leq d$, the exact slice ratio is

$$\frac{\binom{n-2q}{\ell-q}}{\binom{n}{\ell}} = \frac{(\ell)_q (n-\ell)_q}{(n)_{2q}} = \Theta_d((\ell/n)^q), \quad (96)$$

where $(a)_q = a(a-1) \cdots (a-q+1)$. The constants are independent of ℓ : the numerator factors are bounded below by $(\ell/2)^q (n/2)^q$ and above by $\ell^q n^q$, and $(n/2)^{2q} \leq (n)_{2q} \leq n^{2q}$.

Even degree. Suppose $d = 2r$. Use the padded level- ℓ Kikuchi graph: a clause F_i joins S and $S \triangle F_i$ whenever $|S \cap F_i| = r$. The number of ordered instances per original clause is now

$$c_{d,\ell} = \binom{d}{r} \binom{n-d}{\ell-r}.$$

Thus $\bar{d} = c_{d,\ell} W_R / N$, and the comparison identities in Eqn. (29) hold with $c_{d,\ell}$ in place of c . Each label still determines at most one neighbor of a given row, so the weighted Hall allocation and orientation proof is unchanged. Its incoming capacity is $(D+1)\ell$.

By Eqn. (96), $N/c_{d,\ell} = \Theta_d(v^{d/2})$. The sufficient weighted density condition from Eqn. (35) therefore has the form

$$W \geq C'_{d,\epsilon} \ell \frac{N}{c_{d,\ell}}.$$

It is implied by $W \geq C_{d,\epsilon} \mu_d(n, \ell)$ for a sufficiently large constant. With this replacement, the deterministic decomposition, dense-block moments, and spectral certificate from Section 4 apply verbatim. To convert weighted density into label-count density, return the constant certificate W if $W \leq \epsilon M/2$. Otherwise $W > \epsilon M/2$, and a sufficiently large constant in $M \geq C_{d,\epsilon} \mu_d(n, \ell)$ ensures the weighted hypothesis. The weighted certificate then has expectation at most $(41/128)\epsilon W \leq \epsilon M/2$.

Odd degree. Suppose $d \geq 3$ is odd. Retain the constants B, R and the cancellation budgets of Section 6, and replace the bundle thresholds by

$$h_q = \left\lceil BR^q \max\{1, v^{q-d/2}\} \right\rceil, \quad 0 \leq q \leq d-1. \quad (97)$$

The packing proof uses only the exhaustion of earlier stages and $h_{q-1}/h_q \leq 2/R$, so its spread and multiplicity conclusions continue to hold. The number of unbundled labels is at most

$$\frac{nh_{d-1}}{d} = O_{d,\epsilon}(nv^{d/2-1}) = O_{d,\epsilon}(\mu_d(n, \ell)). \quad (98)$$

Fix a positive residual size q and put $a = \lfloor q/2 \rfloor$, $b = \lceil q/2 \rceil$. Each disjoint original pair has

$$T_q = \begin{cases} \frac{1}{2} \binom{q}{a}^2 \binom{n-2q}{\ell-q}, & q \text{ even}, \\ \binom{q}{a}^2 \binom{n-2q}{\ell-q}, & q \text{ odd} \end{cases}$$

balanced undirected instances. Hence $T_q/N = \Theta_d(v^{-q})$. We check that the contention constant remains independent of ℓ . For a competing residual whose intersection with the fixed partner has size $t < a$, the eligibility probability is $O_d(v^{-(a-t)})$. For each fixed intersection, spread bounds the number of competitors by h_{q-t} , with h_q used at $t=0$. Their contribution is at most

$$h_{q-t} O_d(v^{-(a-t)}) \leq O_d(BR^q)(v^{-(a-t)} + v^{b-d/2}) \leq O_d(BR^q).$$

The last inequality uses $b \leq (d-1)/2 < d/2$. If $t \geq a$, then $q-t \leq b < d/2$, so h_{q-t} itself is a constant depending only on d, B, R . This includes $t=q$ and therefore also covers repeated residuals. Summing over the constantly many intersections and four ports gives a uniform contention bound. Exact pair equalization consequently supplies a common pair weight β with

$$\frac{\beta}{N} \geq c_{d,\epsilon} v^{-q}. \quad (99)$$

The good-window orientation still has incoming capacity $(D+1)\ell$. Choose the degree stopping threshold d_* to be the same sufficiently large constant multiple of ℓ used in the trace argument. At a level containing M_q original labels, the direct low-degree remainder is bounded by

$$\begin{aligned} \frac{Nd_*}{\beta M_q h_q} &\leq \frac{C_{d,\epsilon} \ell v^q}{M_q \max\{1, v^{q-d/2}\}} \\ &\leq C_{d,\epsilon} \frac{\ell v^{d/2}}{M_q} = C_{d,\epsilon} \frac{\mu_d(n, \ell)}{M_q}. \end{aligned} \quad (100)$$

Thus a sufficiently large threshold $M_q \geq A_{d,\epsilon} \mu_d(n, \ell)$ pays the same fixed squared error budget as before. Discarding smaller positive levels costs only $O_{d,\epsilon}(\mu_d(n, \ell))$ labels because there are at most $d - 1$ such levels. Diagonal, overlap, extracted-forest, and scalar-block budgets are unchanged. The common-sign certificate assembly of [Section 6](#) now gives the asserted expectation after increasing $C_{d,\epsilon}$.

Cost of the central constructions. For both parities, choose $s = O_{d,\epsilon}(\log N)$ with the same trace thresholds as in the fixed-level proof. Since

$$\ell \log(n/\ell) \leq \log N \leq \ell \log(en/\ell),$$

and $n/\ell \geq 100d$, the memory windows have $L = O_d(\ell + ds) = O_{d,\epsilon}(\log N)$ vertices. Their assignment enumerations cost $N^{O_{d,\epsilon}(1)}$. The trace threshold has inverse polynomial in N , as do the heavy-memory extraction threshold and the required full bias: the normalized traces have Fourier mass at most N , and the local moment potentials have mass $2^{O_{d,\epsilon}(L)}$. All graph dimensions, flow or linear programs, matrix powers, and local certificates therefore cost polynomial time in the explicit input and N . Frozen pair weights and rational arithmetic have polynomial bit length in these same parameters. This proves [Eqn. \(95\)](#) in the central range.

8.2 Boundary levels and scalar derandomization

If $\ell < 2d$ and $n \geq 200d^2$, run the central construction at $\ell' = 2d$. Since $d/2 - 1 \geq 0$, $\mu_d(n, \ell') \leq \mu_d(n, \ell)$, so its density requirement is met. Its polynomial running time fits [Eqn. \(95\)](#).

If $n < 100d\ell$, use the whole-input scalar certificate of [Lemma 9](#), with $u = n$ and $e = \epsilon/2$. The required $M \geq 16e^{-2}(n+1)$ follows from $M \geq C_{d,\epsilon} \mu_d(n, \ell)$ because $\mu_d(n, \ell) \geq n$. Its expectation is at most $\epsilon M/2$ under the specified bias, whose inverse is $2^{O_{d,\epsilon}(n)}$. Construction and evaluation have the same exponential dependence on n , and

$$2^{O_{d,\epsilon}(n)} \leq (en/\ell)^{O_{d,\epsilon}(\ell)}$$

because $n < 100d\ell$ and $\log(en/\ell) \geq 1$. Any case left by these two rules has $n < 200d^2$ and is handled by the same scalar construction with constant overhead depending on d, ϵ . This proves [Proposition 25](#).

For completeness, a scalar signing can also be found directly by conditional expectation of its even-moment potential. This procedure enumerates input assignments, and never enumerates 2^M sign vectors. Fix an input x , put $c_i = a_i \chi_{F_i}(x)$ and $p = 2(n+1)$, and precompute the suffix moments

$$H_j(r) = \mathbb{E} \left(\sum_{i=j}^M c_i \xi_i \right)^r, \quad 0 \leq r \leq p,$$

where the ξ_i are independent uniform signs. Initialize $H_{M+1}(0) = 1$ and $H_{M+1}(r) = 0$ for $r > 0$, and use

$$H_j(r) = \sum_{\substack{0 \leq h \leq r \\ h \text{ even}}} \binom{r}{h} c_j^h H_{j+1}(r-h). \quad (101)$$

If the chosen prefix contributes z at x , fixing the next sign to $b_j = \pm 1$ gives conditional p th moment

$$\sum_{r=0}^p \binom{p}{r} (z \pm c_j)^{p-r} H_{j+1}(r).$$

Sum these expressions over all inputs with the normalization in [Lemma 9](#), and choose a sign whose conditional expectation does not exceed the current one. The uniform starting expectation supplies the desired certificate bound at termination. The recurrence and all conditional comparisons take $\text{poly}(\text{input})2^{O(n)}$ bit operations. The same computation minimizes a sum of finitely many scalar certificates on common signs, since conditional expectations add.

8.3 The avoidance tradeoff

As in [Section 7](#), $n < k$ is handled by evaluating at most 2^k inputs, so assume $n \geq k$ and apply that section's Fourier reduction. If more than n outputs are signed parities, its linear-algebra branch finishes the problem. Otherwise, remove those outputs and retain $M \geq m - n$ nonparity outputs. Their proper Fourier parts have degree at most $k - 1$, with at most $J = 2^k - 1$ layers and Fourier gap $\delta = 2^{1-k}$.

For every layer of degree $2 \leq d \leq k - 1$, use [Proposition 25](#) with $\epsilon = \delta/(4J)$. Since $n/\ell \geq 1$,

$$\mu_d(n, \ell) \leq \mu_{k-1}(n, \ell) = n(n/\ell)^{(k-3)/2}.$$

Degree-zero and degree-one layers use the exact certificates from [Section 7](#). Their expectations are bounded by $\sqrt{M} + \sqrt{\eta}M$ and $\sqrt{nM} + \sqrt{\eta}M$, respectively, so a sufficiently large constant multiple of n and sufficiently small fixed bias give the same margin. Since $\mu_{k-1}(n, \ell) \geq n$, one constant C_k absorbs these requirements and the discarded parity outputs.

Choose a common full bias no larger than any layer requires and enumerate one explicit biased space. The sum of the certificates has expectation at most $\delta M/8$, so some enumerated vector has sum strictly below δM . The contradiction in [Section 7](#) then proves that this vector, with arbitrary values in the discarded positions, is outside the range. There are only $O_k(1)$ layers; taking the smallest of their bias parameters and adding their construction and evaluation costs preserves [Eqn. \(95\)](#). This is precisely [Eqn. \(7\)](#) under [Eqn. \(6\)](#), proving [Theorem 3](#).

9 One-Sided Distinguishers and Data Structure Lower Bounds

Finding one missing string need not distinguish a generator from uniform: the test that accepts just that string has acceptance probability 2^{-m} . The expectation guarantee in [Theorem 2](#) supplies the stronger conclusion needed here. We first retain that guarantee through the Fourier reduction and then apply it to cryptography and data structures. These applications follow the connections developed by Korten, Pitassi, and Impagliazzo [[KPI25](#)] and Guruswami, Lyu, and Yuan [[GLY25](#)]. All locality and accuracy parameters in this section are fixed.

9.1 A Dense, Efficiently Recognizable Subset of the Complement

Theorem 26 (One-sided range test). *For each fixed $k \geq 3$, there are constants $C_k, B_k > 0$ and a deterministic polynomial-time algorithm that, given an explicit k -local map $f : \{0, 1\}^n \rightarrow \{0, 1\}^m$ with $m \geq C_k n^{(k-1)/2}$, constructs a polynomial-size Boolean circuit $D_f : \{0, 1\}^m \rightarrow \{0, 1\}$ satisfying*

$$D_f(f(x)) = 0 \quad \text{for every } x, \quad (102)$$

$$\Pr_{y \sim \mathbf{U}_m} [D_f(y) = 1] \geq \frac{1}{2}. \quad (103)$$

Moreover, for every fully η -biased distribution $\mathcal{Q}$ with $\eta \leq (n + m + 2)^{-B_k}$,

$$\Pr_{y \sim \mathcal{Q}} [D_f(y) = 1] \geq \frac{1}{3}. \quad (104)$$

The circuit D_f is constructed without access to $\mathcal{Q}$.

Proof. Assume first $n \geq k$. If there are more than n signed-parity coordinates, find a nonempty binary dependency among them as in [Section 7](#). Let D_f accept precisely when the corresponding output parity violates the identity obeyed by f . This proves Eqn. (102). Its uniform acceptance probability is $1/2$; under $\mathcal{Q}$ it is at least $(1 - \eta)/2$.

Otherwise retain the $M \geq m - n$ nonparity coordinates. Use exactly the certificates constructed in [Section 7](#), and put

$$T_f(y) = \sum_{A \subseteq [k]} \mathcal{C}_A(y), \quad \delta = 2^{1-k}.$$

Here and below each certificate reads only the retained coordinates of y . Every image point satisfies $T_f(f(x)) \geq \delta M$, while Eqn. (92) gives

$$\mathbb{E}_{\mathcal{Q}} T_f \leq \delta M/8$$

whenever the bias is small enough for every layer. Define $D_f(y) = \mathbf{1}\{T_f(y) < \delta M\}$. Soundness follows from the first inequality, and Markov's inequality gives acceptance probability at least $7/8$ under both uniform and sufficiently small-biased inputs. Construction, exact rational comparison, and Boolean circuit simulation of the evaluation algorithm have polynomial bit complexity.

There is a single exponent B_k that suffices for all instances. Indeed, their truth-table Fourier coefficients have $O_k(1)$ bit length, and the inverse biases in the fixed-parameter constructions are bounded by a polynomial in the explicit representation length $O_k(m \log(n+2) + \log(n+2))$. Increasing B_k absorbs constants and ensures also $\eta \leq 1/3$.

If $n < k$, enumerate the image and let D_f recognize its complement. Uniform acceptance is at least $1 - 2^{n-m} \geq 1/2$. For a fully biased distribution, the character expansion of a singleton gives $\mathcal{Q}(y) \leq 2^{-m} + \eta$. Thus $\mathcal{Q}(\text{Range}(f)) \leq 1/2 + 2^k \eta \leq 2/3$ after a further constant increase of B_k . This computation has polynomial cost at fixed k . $\square$

In the nonparity branch the test can certify a constant Hamming gap. If $T_f(y) \leq \delta M/2$, then

$$\sum_{i=1}^M y_i f_i(x) \leq (1 - \delta)M + T_f(y) \leq (1 - \delta/2)M.$$

Consequently every image differs from y in at least $\delta M/4$ retained positions. This event has probability at least $3/4$. There is no corresponding constant-distance assertion for the single-parity test in the other branch.

Corollary 27 (Any fixed coefficient). *Fix $k \geq 4$ and rational $\rho > 0$. The conclusions of [Theorem 26](#), with constants depending on k, ρ , hold whenever $m > n$ and $m \geq \rho n^{(k-1)/2}$. Construction and evaluation take time*

$$(n + m + 2)^{O_k(1 + \rho^{-2/(k-3)})}.$$

Proof. Choose the constant level $\ell_0 = \max\{1, \lceil (C_k/\rho)^{2/(k-3)} \rceil\}$ in [Theorem 3](#). For $n \geq \ell_0$, its density requirement is at most $\rho n^{(k-1)/2}$. The hierarchy proof preserves the common-bias expectation guarantee before choosing a signing, so the proof of [Theorem 26](#) applies. For $n < \ell_0$, use exhaustive image evaluation and the same singleton bound, increasing the bias exponent by a constant depending on k, ρ . The time bound follows from $\ell_0 = O_k(1 + \rho^{-2/(k-3)})$. $\square$

9.2 Consequences for Local Demi-Bits and Goldreich Generators

We use the convention of Ren, Wang, and Zhong [RWZ26, Definition 2.1]: an (s, ε) -secure demi-bits generator has no nondeterministic circuit of size at most s that rejects its entire image and accepts at least an ε fraction of uniform strings.

Corollary 28 (Breaking local demi-bits). *For every fixed $k \geq 3$, every polynomial-output-length family of k -local maps satisfying $m(n) \geq C_k n^{(k-1)/2}$ admits polynomial-size deterministic distinguishers that reject the entire image and accept at least half of all uniform strings. In particular, such a family is not secure as a demi-bits generator against NP/poly . For $k \geq 4$, the same conclusion holds with any fixed positive coefficient in place of C_k , provided $m > n$.*

Proof. Apply Theorem 26 and corollary 27. A deterministic circuit is a special case of a nondeterministic circuit. For a nonuniform generator family, hardwire its polynomial-length local description and the constructed test. If the local descriptions are uniformly computable, the adversary is uniform as well. $\square$

The attack is one-sided and therefore also distinguishes the output distribution of an ordinary local pseudorandom generator from uniform. If $m \geq rC_k n^{(k-1)/2}$, apply the test to r disjoint output blocks and accept when any test accepts. Uniform blocks are independent, so this raises the acceptance probability to at least $1 - 2^{-r}$; every image point is still rejected. No independence among generator outputs is used.

For Goldreich’s generator, the predicate and the ordered support hypergraph may be arbitrary. For fixed locality $k \geq 4$, a family with output length $m = n^c$ is therefore excluded as a secure demi-bits generator whenever $c \geq (k - 1)/2$. In particular, for the five-variable predicate

$$P(x) = x_1 + x_2 + x_3 + x_4x_5 \quad \text{over } \mathbb{F}_2,$$

the conclusion applies at $m = \rho n^2$ for every fixed $\rho > 0$. This predicate is discussed as a candidate in [RWZ26, Appendix B.3]. The result does not exclude its instantiation at $m = n^{1+\varepsilon}$ for $0 < \varepsilon < 1$. At locality three, the conclusion requires a sufficiently large linear-stretch constant, not every $m > n$ or every specified constant multiple of n .

Constant degree over $\mathbb{F}_2$ is different from constant locality. For example, a quadratic output may involve all input variables. Accordingly, Corollary 28 does not refute the general constant-degree demi-bits assumption in [RWZ26, Assumption 2.3]. It restricts its local candidate instantiations in the parameter range just stated. Li, Ren, and Zhong [LRZ26, Corollary 4.6] already deduce a nondeterministic demi-bits attack at the GLY threshold. Our endpoint removes its logarithmic loss, and Theorem 26 supplies a deterministic one-sided test directly.

The projection barrier to further improvements. Li, Ren, and Zhong [LRZ26, Corollary 4.5] show that, for a universal constant c , a secure k -local demi-bits generator of output length $cs(n)$ rules out even an errorless nondeterministic polynomial-time solver for k -local AVOID at output length $s(n) > n$. Thus the existence of secure local demi-bits at a smaller stretch would obstruct a corresponding improvement of our algorithm. This is a conditional barrier, not an unconditional optimality claim for the exponent $(k - 1)/2$.

Proof complexity. There is also a direct consequence at the level of general propositional proof systems. For each fixed k , extend a standard complete proof system by a rule that accepts a range-exclusion tautology for (f, y) when the promise of Theorem 26 holds and $D_f(y) = 1$. The rule is sound and polynomial-time checkable. It gives polynomial-size proofs for at least half of the

strings y at the stated stretch. This defines a Cook–Reckhow proof system; it is not a claim of short proofs in Resolution, Polynomial Calculus, or a fixed-degree sum-of-squares system. Establishing any such claim requires a separate simulation of the certificate construction.

The projection argument of Li, Ren, and Zhong [LRZ26, Theorem 3.2 and Section 4] strengthens the uniform-output guarantee. We record its quantitative consequence; the amplification mechanism is theirs.

Corollary 29 (Short certificates for almost all candidate outputs). *Fix $k \geq 3$, and let s, m be integers with*

$$n < s \leq m, \quad s \geq C_k n^{(k-1)/2}.$$

There is a polynomial-time verifier $V_k(f, y, I)$, uniform in the local description of $f : \{0, 1\}^n \rightarrow \{0, 1\}^m$, whose witness I is an s -element subset of $[m]$, such that

1. $V_k(f, y, I) = 1$ implies $y \notin \text{Range}(f)$;
2. the number of strings having no accepting witness is at most

$$\sum_{j=0}^{s-1} \binom{m}{j}. \tag{105}$$

In particular, if $m \geq 4s$, the verifier accepts, for some witness, at least a $1 - 2^{-\gamma m}$ fraction of uniform strings, where $\gamma = 1 - h_2(1/4) > 0$ and $h_2(u) = -u \log_2 u - (1 - u) \log_2(1 - u)$. For each fixed k , one Cook–Reckhow proof system gives polynomial-size proofs of the corresponding range-exclusion tautologies for all these accepted strings.

Proof. Let A_k be the deterministic algorithm of Theorem 1. The verifier first rejects malformed local descriptions and witnesses that violate the stated length and density conditions. Given I , form the projected map $f|_I$ in increasing coordinate order, compute $z_I = A_k(f|_I)$, and accept precisely when $y|_I = z_I$. Projection preserves locality and the density promise holds at output length s . Thus z_I is outside the range of $f|_I$, proving soundness. An m -bit incidence vector encodes I , and verification takes polynomial time for fixed k .

Let U be the set of strings with no accepting witness. For every s -element set I , the projection $U|_I$ omits z_I . Consequently U shatters no set of s coordinates. The Sauer–Shelah bound, recalled in [LRZ26, Lemma 2.10], gives Eqn. (105). If $s \leq m/4$, the binomial entropy bound gives

$$2^{-m}|U| \leq 2^{-m} \sum_{j=0}^{s-1} \binom{m}{j} \leq 2^{-m(1-h_2(1/4))}.$$

Finally, extend a fixed complete Cook–Reckhow system by a proof format containing (f, y, I) , checked by V_k , and producing the canonical range-exclusion tautology for (f, y) . The verifier is sound and polynomial-time, and the base system preserves completeness. Its proof lengths are polynomial in the local description and output length. $\square$

More generally, $s \leq \beta m$ for fixed $0 < \beta < 1/2$ gives exceptional fraction at most $2^{-(1-h_2(\beta))m}$. At $s = \lceil C_k n^{(k-1)/2} \rceil$, increasing C_k if needed to ensure $s > n$, the condition $m \geq 4s$ changes only the constant in the main stretch bound. This conclusion uses the AVOID algorithm alone. Finding a witness I for a given y is not shown to be deterministic polynomial time, and the exponentially small exceptional fraction is asserted for uniform y , not for every small-biased distribution. The deterministic test and small-bias guarantee of Theorem 26 remain separate inputs to the data

structure applications. Likewise, the proof system constructed here does not imply short proofs in an independently specified weak system. Transferring the result to such a system requires short proofs for the projected instances and a suitable closure under adding axioms; see [LRZ26, Section 4.3].

These are average-case guarantees over candidate outputs of each fixed map. They do not constitute a worst-case-to-average-case reduction over a fixed distribution of circuit instances. The conditional average-case hardness result in [LRZ26, Theorem 4.4] instead starts from a demi-bits generator and samples its output projections.

9.3 Universal Families and Nonadaptive Bit Probes

Corollary 30 (An oblivious family of candidate nonoutputs). *For fixed $k \geq 3$ and lengths n, m satisfying the density promise, there is an explicitly enumerable multiset $\mathcal{H}_{n,m}$ of $(n+m)^{O_k(1)}$ strings such that, for every k -local map $f : \{0,1\}^n \rightarrow \{0,1\}^m$, at least one third of the elements of $\mathcal{H}_{n,m}$, counted with multiplicity, are accepted by D_f . The multiset is chosen before f is specified.*

Proof. Use Lemma 6 with $h = m$ and $\eta = (n+m+2)^{-B_k}$, and apply Eqn. (104). $\square$

The quantifiers are $\exists \mathcal{H} \forall f \exists y \in \mathcal{H}$, not $\exists y \forall f$. They are exactly what the data structure application needs. A fixed query algorithm must represent every row of the query problem, although each row has its own memory encoding.

A static query problem is a Boolean matrix $F : [R] \times [m] \rightarrow \{0,1\}$. An encoding assigns each row a memory string of S bits. A nonadaptive t -probe query algorithm reads at most t memory locations, determined by the query index, and returns the requested entry. No computational bound is placed on encoding or decoding.

Lemma 31 (Strongly explicit rows). *For every fixed integer $B \geq 1$ and $m \geq 1$, a fully $(3m+3)^{-B}$ -biased distribution on m bits can be indexed by $O_B(\log(m+2))$ uniform seed bits. Any specified output coordinate is computable deterministically in time $\text{poly}_B(\log(m+2))$ from the seed, the coordinate index, and m .*

Proof. Put $\eta = (3m+3)^{-B}$ and choose r to be the least power of two at least $\lceil \log_2 \max\{2, m/\eta\} \rceil$. Use the construction in Lemma 6 over $\mathbb{F}_{2^r}$, indexed by $(a, z) \in \mathbb{F}_{2^r}^2$. Its bias is at most $(m-1)/2^r \leq \eta$, and the seed length is $2r = O_B(\log(m+2))$.

For completeness, this field has a deterministic quadratic-tower construction of cost polynomial in r . Begin with $K = \mathbb{F}_2$ and $c = 1$, whose absolute trace is one. Replace K by $K[u]/(u^2 + u + c)$. The polynomial is irreducible because $\text{Tr}_{K/\mathbb{F}_2}(c) = 1$: the image of $v \mapsto v^2 + v$ is exactly the trace-zero subspace. In the new field, cu has absolute trace one, since its relative trace to K is c . Use cu in the next extension. Repeating until dimension r gives arithmetic in a specified binary basis in time polynomial in r . Computing z^{j-1} and its trace then has cost polynomial in $r + \log m$. This avoids any need for an exponential search for an irreducible polynomial when evaluating a query. $\square$

Theorem 32 (Logarithm-free nonadaptive space bound). *For every fixed $t \geq 3$, there is $B_t > 0$ such that the following holds. Let the rows of $F : [R] \times [m] \rightarrow \{0,1\}$ support a fully $(3m+3)^{-B_t}$ -biased distribution. Every exact nonadaptive t -bit-probe representation of F requires*

$$S \geq c_t m^{2/(t-1)} \quad (106)$$

for a constant $c_t > 0$. There are explicit such matrices with $R = m^{O_t(1)}$ and entries computable in time $\text{poly}_t(\log(m+2))$.

Proof. If $S \geq m$, Eqn. (106) is immediate after taking $c_t \leq 1$. Otherwise a fixed decoder induces the t -local map

$$C : \{0, 1\}^S \longrightarrow \{0, 1\}^m, \quad C(z)_j = \text{the answer to query } j \text{ on memory } z.$$

Correctness for every row implies $\text{supp}(\mathcal{Q}) \subseteq \text{Range}(C)$. If $m \geq C_t S^{(t-1)/2}$, Theorem 26 contradicts this containment: the chosen bias is at most $(S + m + 2)^{-Bt}$ and the test accepts a positive fraction of $\mathcal{Q}$ outside the range. Hence $m < C_t S^{(t-1)/2}$, which gives the bound. Zero-bit memories are impossible for a distribution with bias less than one.

For explicitness, use Lemma 31 on m bits at the stated bias. Its sample points are the rows, with multiplicity. The seed length is $O_t(\log(m + 2))$ and its coordinate evaluation has the asserted running time. $\square$

In particular, $t = 3$ gives $S = \Omega(m)$, while $t = 4$ gives $S = \Omega(m^{2/3})$ and $t = 5$ gives $S = \Omega(m^{1/2})$. The row index has only $O_t(\log m)$ bits. The three-probe statement is therefore an explicit linear-space lower bound for a problem whose data can be specified with logarithmically many bits. The earlier reduction in [GLY25, Theorem 10] gives the same power with a logarithmic loss. Our statement removes that loss for a sufficiently small polynomial bias; we do not preserve their specific bias exponent.

9.4 Adaptive Queries and Constant-Size Words

We give a separate consequence for adaptive cell probes. It uses the full degree- t expansion, rather than treating a depth- t decision tree as a junta on $2^t - 1$ bits. This distinction avoids an exponential loss in the exponent of the storage bound.

Theorem 33 (Remote points for bounded-depth word queries). *Fix integers $t \geq 2, w \geq 1$ and rational $0 < \varepsilon < 1/2$. There are constants $C, B > 0$ depending on these parameters such that, for integers $S, m \geq 1$, if each coordinate of $f : (\{0, 1\}^w)^S \rightarrow \{\pm 1\}^m$ is computed by an adaptive decision tree making at most t word queries and $m \geq CS^{t/2}$, a deterministic polynomial-time test certifies*

$$\min_z \Delta(f(z), y) \geq \frac{1}{2} - \varepsilon$$

with probability at least $3/4$ under every fully $(S + m + 2)^{-B}$ -biased distribution on y . The test takes the explicit query trees as input and never certifies a false assertion.

Proof. For every cell v and nonzero $a \in \mathbb{F}_2^w$, introduce the formal sign variable $X_{v,a} = (-1)^{\langle a, z_v \rangle}$. There are $N = (2^w - 1)S$ such features. Delete repeated queries to an already known cell along any path of a decision tree. The indicator of a leaf is a product of at most t word-value indicators, and

$$\mathbf{1}\{z_v = c\} = 2^{-w} \sum_{a \in \mathbb{F}_2^w} (-1)^{\langle a, c \rangle} X_{v,a}, \quad X_{v,0} = 1.$$

The resulting Fourier expansion has degree at most t in the formal features and at most $J = 2^{2wt}$ nonzero monomials per output. After collecting terms, every coefficient has magnitude at most one: these are Fourier coefficients of a Boolean function on the original independent uniform cells. Each monomial uses at most one feature from any cell.

Order the monomials separately by degree in each output, pad lists with zero coefficients, and form at most $(t + 1)J$ systems, with one original label per output in every system. For degrees at least two, apply Theorem 2 on the N formal variables. Treating the features as independent enlarges

the domain and hence preserves the upper-bound direction. Degrees zero and one use [Lemma 23](#). Pad the formal variable set to size at least t if necessary; this changes only fixed constants.

Choose each accuracy parameter as $\varepsilon/((t+1)J)$. The density hypothesis, with sufficiently large C , satisfies every layer's requirement. Their sum $T(y)$ is nonnegative, efficiently evaluable, and obeys

$$\sup_z \left| \sum_i y_i f_i(z) \right| \leq T(y), \quad \mathbb{E}T \leq \varepsilon m/2.$$

All inverse biases are uniformly polynomial in $S+m$ for fixed t, w, ε . Markov's inequality gives $T(y) \leq 2\varepsilon m$ with probability at least $3/4$. The identity $\Delta(f(z), y) = (1 - m^{-1} \sum_i y_i f_i(z))/2$ completes the proof. $\square$

Corollary 34 (Adaptive constant-word space bound). *For fixed $t \geq 2, w \geq 1$, there is an explicit query problem with $m^{O_{t,w}(1)}$ rows such that every exact adaptive t -probe representation using w -bit cells requires*

$$S = \Omega_{t,w}(m^{2/t}).$$

The same bound holds if each row need only be represented within relative Hamming error $1/4$.

Proof. Take $\varepsilon = 1/8$ in [Theorem 33](#) and choose a common bias $(3m+3)^{-B}$ for all $S < m$. Use [Lemma 31](#). Its explicit small-biased rows cannot all lie within distance $1/4$ of the decoder's range, because a positive fraction have distance at least $3/8$. The case $S \geq m$ is immediate. A zero-cell decoder is constant; its average distance from a small-biased row is at least $(1-\eta)/2 > 1/4$, so it cannot provide the stated approximation. $\square$

The word size is fixed: this corollary makes no assertion of uniform dependence on a growing $w = \Theta(\log m)$. Also, when $w = 1$, the specialized adaptive bit-probe analysis of [\[GLY25, Section 5.2\]](#) has stronger powers of m than $2/t$. Thus [Corollary 34](#) is a logarithm-free baseline and a robust approximation statement, not a replacement for those stronger adaptive bit-probe results. All data structure assertions above concern exact deterministic decoding, or the explicitly stated per-row Hamming approximation; bounded-error randomized decoding requires a separate argument.

10 What the Sharp Moore Bound Supplies Directly

The sharp density scale in our construction comes from the recent proofs of the hypergraph Moore bound [\[BKNN⁺26, SH26b\]](#). Bandeira et al. [\[BKNN⁺26, Theorem 1.2, version 2\]](#) prove that, for every fixed uniformity $d \geq 3$, density $M \geq c_d n(n/\ell)^{d/2-1}$ forces an even cover of size at most $C_d \ell \log n$. Their second version covers odd as well as even d . Schmidhuber and Hastings additionally state the refined cover-length bound $O(\ell \log(en/\ell))$ with absolute constants. Their random-instance companion [\[SH26a\]](#) does not by itself supply our arbitrary-support, common-sign certificate theorem. This section records a direct consequence of the extremal result and separates it from the constructive work.

Proposition 35 (Many disjoint short dependencies). *Fix $d \geq 3$ and $1 \leq \ell \leq n$. There are constants $A, C > 0$ such that every labeled d -uniform hypergraph with*

$$M \geq 2R, \quad R = \left\lceil Cn(n/\ell)^{d/2-1} \right\rceil$$

has a collection of pairwise label-disjoint nonempty even covers, each of size at most $L = \max\{2, \lceil A\ell \log(en/\ell) \rceil\}$, whose union contains at least $M/2$ labels.

Proof. While at least R labels remain, find an even cover and delete it. Two repeated supports immediately give a cover of size two; otherwise apply the sharp Moore bound to the simple residual hypergraph. Choose C large enough for its strict density inequality. Fewer than R labels remain at termination, so at least $M/2$ have been removed. $\square$

Corollary 36 (A weak refutation margin from short covers). *Under the hypotheses of Proposition 35, consider the signed equations $\chi_{F_i}(x) = b_i$. Under every fully η -biased distribution on the M labels, with probability at least $1 - 8L/M - 4\eta$, every assignment violates at least $M/(8L)$ equations. Equivalently, the maximum normalized correlation is at most $1 - 1/(4L)$.*

Proof. Fix the packing before drawing the signs, and let it have q covers $E_1, \dots, E_q$. Since their union has at least $M/2$ labels, $q \geq M/(2L)$. Put $Z_j = \prod_{i \in E_j} b_i$. Every $Z_j = -1$ certifies an inconsistent subsystem: multiplying its equations has left-hand side one. The covers are label-disjoint, so every $Z_j Z_h$ with $j \neq h$ is a nonempty parity. Therefore

$$\mathbb{E} \left(\sum_{j=1}^q Z_j \right)^2 \leq q + \eta q(q-1).$$

Markov's inequality bounds the probability that $\sum_j Z_j > q/2$ by $4/q + 4\eta \leq 8L/M + 4\eta$. On the complementary event, at least $q/4$ covers are inconsistent. Each forces a violation on a different label, proving the claim. $\square$

The covers provide polynomial-size verifiable witnesses even when this existence proof does not find them in polynomial time at a growing level. The weak margin is $\Theta(1/L)$, not a fixed constant. Passing from these witnesses to efficiently constructed strong cancellation for arbitrary supports is precisely why the memory extraction, pair equalization, and forest decomposition in the preceding sections are needed. Neither the short-cover theorem alone nor the present fixed-accuracy analysis proves a smaller polynomial-time stretch exponent or automatically removes every logarithmic factor from the specialized adaptive-query bounds.

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